

Tracking Evolving Exploit Discourse in Hacker Communities: A Diachronic Exploit Text Graph Framework for Proactive Cyber Threat Intelligence

*Cybersecurity threats emerge and evolve in hacker communities long before they are formally catalogued in vulnerability databases, yet existing threat-intelligence systems process static snapshots of hacker discourse, missing the temporal semantic drift that signals emerging attack patterns. We present the **Exploit Text Graph (ETG)**—a cumulative temporal co-occurrence graph of exploit vocabulary—and the **Diachronic Graph Transformer (DGT)**, an adjacency-masked graph transformer that embeds ETG nodes across successive temporal spells and predicts which terms will undergo the greatest semantic shift in the next observation window. Applied to HackerSignal, a 1,783,276-post multi-source corpus spanning 12 temporal spells from January 2016 through April 2026, DGT achieves a heldout mean absolute error of 0.082 (a 25% improvement over a naive mean baseline) and a Spearman ρ of 0.662 on next-spell shift prediction, significantly outperforming all 36 of 36 tested external benchmark models—including GloVe, PPMI, SGNS, and five dynamic graph neural network baselines—at a false-discovery-rate-corrected $q < 0.05$. Rank-based evaluation via Spearman ρ is particularly discriminating: null-predictor models that achieve low MAE by predicting zero shift for all terms show $\rho < 0$, confirming they actively rank high-shift terms last and are counter-productive for watchlist generation. Ablation experiments validate four design principles: cumulative*

graph encoding, adjacency-constrained self-attention with Laplacian positional encodings, a recency-weighted temporal context loss, and multi-source corpus construction. A six-term case study traces semantically pivotal cybersecurity events—from the Mirai botnet and WannaCry through the emergence of Ransomware-as-a-Service and LLM jailbreaking—demonstrating that DGT detects vocabulary realignment in real time and produces an actionable forward-looking watchlist of predicted next-spell shifts. Theoretical contributions, design principles, and practitioner implications for proactive cyber threat intelligence are discussed.

Keywords: cyber threat intelligence, design science research, diachronic graph transformer, exploit text graph, hacker forum analytics, temporal semantic shift

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1 INTRODUCTION

The cybersecurity threat landscape is defined by an asymmetric detection lag: adversaries develop, refine, and share novel attack techniques in hacker forums long before those techniques enter the formal record of known vulnerabilities. By the time a vulnerability receives a Common Vulnerabilities and Exposures (CVE) identifier and appears in catalogues such as the CISA Known Exploited Vulnerabilities (KEV) list, security teams face a compressed remediation window—often measured in days (CISA, 2024; Jacobs et al., 2021). Our analysis of 1,783,276 hacker-forum posts collected between January 2016 and April 2026 reveals that 271,275 distinct CVE identifiers were mentioned in hacker discourse, reflecting a vast pool of vulnerability intelligence that circulates in underground communities well before, during, and after formal classification. Bridging this information gap is a central challenge for proactive cyber threat intelligence (CTI).

Hacker communities are early-warning systems for emerging threats. Prior research in IS cybersecurity analytics has demonstrated that dark-web forums, hacker boards, and underground marketplaces contain signals predictive of real-world exploit activity (Ebrahimi et al., 2022; Nunes et al., 2016; Samtani et al., 2017, 2022). These communities develop, contest, and evolve a shared technical vocabulary—a lexicon in which terms not only denote artifacts (tools, vulnerabilities, techniques) but also encode the community’s collective understanding of how those artifacts relate to one another. The vocabulary itself is dynamic: the word *jailbreak* referred exclusively to iOS

firmware bypass through 2022 before migrating, almost entirely, to large-language-model (LLM) safety-filter circumvention following the public release of ChatGPT. The word *stealer* shifted from a generic adjective to a recognised product category (e.g., RedLine Stealer, Raccoon Stealer) with versioned releases and subscription pricing. Tracking these *semantic shifts*—changes in how terms are used relative to their co-occurrence neighbours—offers a mechanism for detecting emerging attack paradigms before they crystallise into named CVEs.

Despite this potential, existing IS cybersecurity analytics systems treat hacker discourse as a static or bag-of-words phenomenon. Text-mining methods (Samtani et al., 2017; Tavabi et al., 2018), deep transfer learning (Samtani et al., 2022), and cross-lingual models (Ebrahimi et al., 2022) extract threat signals from individual snapshots or aggregate corpora, but none model the continuous temporal drift of exploit vocabulary as a predictive signal. Classical diachronic semantic models (Hamilton et al., 2016; Kim et al., 2014) capture word-level drift but discard the relational co-occurrence structure that encodes how exploit techniques combine—a structure central to understanding attack chains. Dynamic graph neural networks (Rossi et al., 2020; Sankar et al., 2020) model temporal graph evolution but are not designed for the shift-prediction task or for the cumulative nature of exploit knowledge. The result is a methodological gap: no prior IS artifact simultaneously (1) represents exploit discourse as a relational graph, (2) learns diachronic embeddings across sequential graph snapshots, and (3) forecasts next-period semantic shift to support proactive CTI.

We address this gap through a Design Science Research (Hevner et al., 2004; Peffers et al., 2007) programme that produces two interlocking IS artifacts. The **Exploit Text Graph (ETG)** is a cumulative directed weighted graph in which nodes represent stemmed exploit terms and edges represent co-occurrence within a four-token sliding window across all posts up to and including

each temporal spell. The ETG grows monotonically—capturing the accumulation of community knowledge—and is constructed at 12 spell boundaries spanning a decade of hacker discourse. The **Diachronic Graph Transformer (DGT)** is an adjacency-constrained graph-attention encoder trained on the sequence of ETG snapshots to produce per-term embeddings whose cosine distance between consecutive spells quantifies semantic shift. Next-spell shift forecasting is performed post-hoc—after training—by applying EMA extrapolation to the per-spell embedding trajectories, enabling the construction of a forward-looking watchlist of terms expected to undergo the greatest semantic realignment.

This paper makes three primary contributions to IS research and practice. First, we contribute two novel IS artifacts—the ETG and DGT—accompanied by four empirically validated design principles. Second, we contribute a rigorous 38-model benchmark evaluation spanning non-neural diachronic baselines, static and dynamic graph embeddings, and modern graph transformers, with paired significance testing and false-discovery-rate correction. Third, we contribute a six-term case study grounded in publicly verifiable cybersecurity events, demonstrating that DGT correctly identifies semantic pivots associated with the Mirai botnet, WannaCry/EternalBlue, Ransomware-as-a-Service emergence, info-stealer marketplaces, Ivanti zero-days, and LLM jailbreaking—each a shift that a practitioner would recognise as a meaningful change in the threat landscape.

The remainder of this paper is organised as follows. Section 2 reviews related work across four theoretical streams. Section 3 describes the Design Science Research approach. Section 4 presents the artifact design. Section 5 reports the benchmark evaluation. Section 6 presents the case study. Section 7 discusses theoretical contributions, practical implications, and limitations. Section 8 concludes.

2 THEORETICAL BACKGROUND AND RELATED WORK

2.1 IS Cybersecurity Analytics and Hacker Community Intelligence

Hacker communities—including dark-web forums, surface-web hacker boards, and underground marketplaces—have become a primary data source for proactive cybersecurity analytics in the IS literature. Foundational work by Samtani et al. (2017) introduced a hacker asset framework that identifies emerging tools, exploits, and key actors in hacker forums, demonstrating that community-level text analytics can surface intelligence unavailable from traditional vulnerability feeds. Subsequent work extended this framework to link dark-web exploit postings to known CVEs using semantic embedding techniques, showing that deep representations of hacker posts can predict exploitation activity (Samtani et al., 2022). Cross-lingual extensions (Ebrahimi et al., 2022) addressed the multilingual character of global hacker communities, applying zero-shot and adversarial learning methods to non-English forums, while vulnerability prioritisation research (Ullman & Samtani, 2024) demonstrated that multi-view learning over vulnerability metadata improves CVSS scoring and KEV classification.

A parallel literature focuses on SCADA and IoT vulnerability intelligence, extracting structured threat indicators from unstructured hacker posts (Samtani et al., 2020), and on the epidemiology of exploit diffusion through underground networks (Nunes et al., 2016; Tavabi et al., 2018). Collectively, this body of work establishes that hacker communities are rich, timely, and idiosyncratic sources of threat intelligence—but it also reveals a shared limitation: all existing IS cybersecurity analytics methods process hacker discourse at a single point in time or aggregate across time, treating the corpus as a static collection. None model how the *meaning* of exploit terms evolves continuously, nor do they forecast which terms are approaching a semantic transition. Our

work directly addresses this gap.

2.2 Diachronic Semantic Models

Diachronic (time-varying) semantic models seek to track how the meaning of words changes across corpora collected at different time periods. The foundational approach of Hamilton et al. (2016) trains Word2Vec embeddings (Mikolov et al., 2013) on successive temporal corpora and aligns them using Procrustes rotation, yielding cosine distances that measure per-word semantic change. Kim et al. (2014) extended this approach with temporal neural language models that explicitly condition word representations on time. Kulkarni et al. (2015) proposed statistical tests for detecting significant linguistic change in large corpora. More recently, Röttger and Pierrehumbert (2021) demonstrated that fine-tuning BERT (Devlin et al., 2019) on time-stamped data improves temporal adaptation on downstream tasks, though at the cost of retraining for each new time period.

These methods share a fundamental limitation for the CTI application: they operate at the word level and discard the relational structure of co-occurrence. In hacker discourse, meaning is conveyed not only by individual terms but by the combinations of terms that co-occur—*ransomware* and *affiliate* co-occurring with *panel* signals a different context than *ransomware* co-occurring with *decryption*. A graph representation captures this combinatorial semantics explicitly, enabling the model to detect when a term’s neighbourhood changes—which is, precisely, what semantic shift means in a community of practice. Furthermore, diachronic word embedding methods are designed to measure shift, not to *predict* it. Our DGT produces per-spell embeddings from which next-spell cosine shift is forecast post-hoc via EMA extrapolation, enabling proactive rather than retrospective analysis.

2.3 Graph Neural Networks and Dynamic Graph Learning

Graph neural networks (GNNs) generalise deep learning to graph-structured data by propagating information across node neighbourhoods (Hamilton et al., 2017; Kipf & Welling, 2017; Veličković et al., 2018). Graph embedding methods—DeepWalk (Perozzi et al., 2014), Node2Vec (Grover & Leskovec, 2016), LINE (Tang et al., 2015)—learn static node representations by factorising adjacency or applying random walks, providing strong baselines for co-occurrence graph representation. For temporal graphs, dynamic graph neural networks model evolving node states: DySAT (Sankar et al., 2020) applies self-attention across structural and temporal dimensions; EvolveGCN (Pareja et al., 2020) uses recurrent units to evolve GCN weight matrices; TGN (Rossi et al., 2020) combines memory modules with graph attention; and JODIE (Kumar et al., 2019) projects user and item trajectories through time. These models have been applied to link prediction and node classification on interaction graphs, but not to the diachronic co-occurrence graph setting or the shift-prediction objective.

Graph Transformers adapt the self-attention mechanism (Vaswani et al., 2017) to graph inputs. GraphGPS (Rampášek et al., 2022) integrates local MPNN layers with global attention and Laplacian positional encodings (Kreuzer et al., 2021). TokenGT (Kim et al., 2022) treats graphs as sequences of node and edge tokens, enabling full quadratic attention. These architectures excel on molecular property prediction and graph classification benchmarks, but have not been applied to temporal co-occurrence graphs in the NLP or security domain. Our DGT adapts the graph transformer paradigm specifically for diachronic exploit graph embedding by introducing adjacency-constrained within-spell attention—restricting self-attention to ETG co-occurrence neighbours so that each node’s representation encodes relational graph semantics

rather than global term proximity—combined with a recency-weighted temporal context loss that enforces temporally ordered embedding trajectories. Post-hoc EMA extrapolation over the resulting per-spell embeddings then yields a shift forecast absent from all prior graph transformer formulations.

2.4 The ETG and DGT Paradigm: Filling the Gap

Table 1: Comparison of Related Work on Key Dimensions					
Approach	Representative Works	IS Domain	Graph	Temporal	Shift Pred.
IS CTI Analytics	Samtani et al. (2017, 2022)	✓	×	×	×
Diachronic Semantics	Hamilton et al. (2016); Kim et al. (2014)	×	×	✓	×
Dynamic GNNs	Rossi et al. (2020); Sankar et al. (2020)	×	✓	✓	×
Graph Transformers	Rampášek et al. (2022); Kim et al. (2022)	×	✓	×	×
ETG + DGT (ours)	This paper	✓	✓	✓	✓
<i>Note: ✓ = supported; × = not supported; ~ = partially supported. IS Domain refers to hacker-community or cybersecurity analytics; Graph Structure to relational co-occurrence representation; Temporal to sequential spell modelling; Shift Prediction to forecasting next-period semantic change.</i>					

As shown in Table 1, no prior work occupies all four dimensions simultaneously. The ETG fills the relational IS-domain gap by constructing a cumulative graph of exploit co-occurrences grounded in hacker forum data. The DGT fills the temporal shift-prediction gap by applying adjacency-constrained graph-attention within each ETG snapshot, training with a recency-weighted temporal context loss to enforce diachronically ordered embedding trajectories, and forecasting next-period cosine drift post-hoc via EMA extrapolation. Together, they instantiate four design principles derived from this gap analysis:

- **DP1 — Cumulative Encoding:** Exploit knowledge accumulates; each spell’s graph should encode all prior co-occurrence evidence, not only the current period.

- **DP2 — Adjacency-Constrained Self-Attention:** Attention between nodes is restricted to observed ETG co-occurrence neighbours, encoding relational exploit structure. Nodes additionally carry Laplacian spectral positional encodings that ground representations in graph topology. Temporal isolation is enforced architecturally through per-spell independent encoding.
- **DP3 — Recency-Weighted Temporal Context Loss:** The training objective includes a novel EMA-based regulariser encouraging temporal continuity in embedding space, pulling current-spell embeddings toward a smooth exponential average of past spell embeddings so that diachronic trajectories remain ordered and smooth across spells.
- **DP4 — Multi-Source Corpus:** Threat intelligence requires breadth; the corpus must combine hacker communities, vulnerability references, advisories, exploit archives, and related sources.

3 DESIGN SCIENCE RESEARCH APPROACH

3.1 DSR Methodology

We adopt the Design Science Research methodology of Hevner et al. (2004), which frames IS research as the creation and evaluation of purposeful artifacts that address organisational problems. The three DSR cycles—relevance, design, and rigor—structure our work. The relevance cycle grounds the research in the practitioner need for proactive CTI. The design cycle encompasses the construction and instantiation of the ETG and DGT artifacts. The rigor cycle connects the artifacts to the prior knowledge base in diachronic semantics, graph learning, and IS cybersecurity analytics. We additionally apply the six-step DSRM process of Peffers et al. (2007): problem identification,

objectives of a solution, design and development, demonstration, evaluation, and communication (Gregor & Hevner, 2013).

3.2 Problem Relevance

The practitioner problem is well-documented: CTI analysts at security operations centres, CISOs at enterprise organisations, and analysts at Information Sharing and Analysis Centres (ISACs) require timely intelligence about emerging threats. Existing tools—SIEM platforms, signature-based intrusion detection, static vulnerability feeds—are reactive; they alert after an attack technique is known and catalogued. Hacker forums contain this information earlier, but no IS artifact operationalises the temporal dimension of that intelligence. The problem is an IS problem—not a purely technical one—because it involves the design of an artifact for a knowledge-work context (CTI analysis), requires organisational data (multi-source hacker corpora), and has implications for managerial decision-making (vulnerability prioritisation, resource allocation for patching).

3.3 Design Objectives

Three design objectives flow directly from the problem and the gap analysis in Section 2.4. First, the artifact must *represent exploit relationships* as a graph, capturing how terms co-occur and combine. Second, it must *model temporal drift* across successive observation periods, learning embeddings that encode semantic evolution. Third, it must *generate actionable predictions*—a ranked watchlist of terms predicted to shift meaning in the next period—that a CTI analyst can operationalise without additional modelling expertise. These objectives map directly to the ETG, the DGT encoder, and the watchlist generation procedure, respectively.

3.4 Evaluation Strategy

Following Hevner et al. (2004)’s taxonomy of evaluation modes, we employ two complementary approaches. First, a *technical evaluation* benchmarks DGT against 38 models across four experiment tiers—non-neural diachronic baselines, graph embedding and dynamic GNN baselines, modern graph transformers, and DGT component ablations—with paired significance testing and Benjamini–Hochberg false discovery rate correction (Benjamini & Hochberg, 1995). This evaluation addresses the question of whether DGT advances the state of the art on the shift-prediction task. Second, a *naturalistic evaluation* presents a six-term case study anchored to publicly verifiable real-world cybersecurity events, demonstrating the artifact’s practical utility without requiring a controlled laboratory study. A formal user study with CTI practitioners is identified as priority future work.

4 ARTIFACT DESIGN

4.1 HackerSignal Corpus

4.1.1 The Full HackerSignal Dataset

The ETG analysis draws on HackerSignal (Ampel & Samtani, 2026), a large-scale multi-source benchmark corpus linking hacker community discourse to the CVE vulnerability lifecycle. The full HackerSignal release comprises **7,447,646 exact-deduplicated documents** from **64 public forum and source identifiers** spanning **eight source layers** and a **36-year window** (1991–2026):

1. **Hacker Community (6,966,014 docs; 56 forums):** Organic security discourse from darkweb marketplaces, security discussion boards, and multilingual hacker communities. Major forums include `gayanku_darkweb` (3.11M rows, 2011–17), `antichat`

(1.82M, 2002–26), `evolution_darkweb` (471K, 2014–15), `antionline` (463K, 2001–26), `persiantools` (280K), `wwhclub` (161K), and `hackforums` (151K, 2009–15), alongside 48 additional forums.

2. **Vulnerability Reference (340,536 docs):** National Vulnerability Database (NVD) CVE descriptions (2002–26), each with a CVE identifier; the retrieval corpus for HackerSignal’s benchmark tasks.
3. **Exploit Archive (46,653 docs):** ExploitDB, 0day.today, PacketStorm, ZeroScience, and Vulnerability-Lab proof-of-concept exploit descriptions (1991–2026).
4. **Exploit Q&A Reference (46,505 docs):** Structured exploit question-and-answer references (circa 2000).
5. **Advisory Reference (28,684 docs):** GitHub Advisory, CISA advisories, and CERT notices (2017–2026).
6. **Bug-Bounty Disclosure (9,453 docs):** HackerOne and related bug-bounty platform reports (2013–2024).
7. **Fix-Commit Reference (8,293 docs):** CVEfixes security-relevant fix commit messages (1999–2022).
8. **Exploitation Reference (1,508 docs):** CISA Known Exploited Vulnerabilities (KEV) catalog entries (2021–2026).

Documents are unified through a shared CVE identifier namespace and a common JSONL record format carrying source provenance, ISO 8601 timestamps, and pseudonymized author

hashes. Across all 7.45M documents, 4.8% carry explicit CVE references, 23.5% contain at least one CTI keyword, and estimated near-duplicates account for 8.2% (addressed through SHA-256 exact deduplication and MinHash-LSH near-deduplication during corpus construction). The corpus spans multilingual content—including Russian, Turkish, Persian, Arabic, and Chinese in DeepDarkCTI and darknet sources—though English-language posts constitute the substantial majority.

4.1.2 ETG Corpus: Subsetting HackerSignal to the Analysis Window

The ETG analysis requires a *temporally coherent* corpus aligned with the modern era of systematic CVE tracking and structured exploit discourse. We apply a single temporal filter: **January 1, 2016 onward**. The 2016 boundary is motivated by three converging factors. First, the CISA KEV catalog, the NVD’s modern structured CVE format, and the major hacker community platforms that dominate our analysis all reach maturity and consistent coverage after 2016. Second, the pre-2016 period is dominated by a small number of historical forum archives (Gayanku 2011–17, Evolution 2014–15) whose temporal coverage is asymmetric with respect to the other source layers. Third, the ETG’s design is oriented toward the *modern* exploit discourse era—the decade in which IoT weaponisation, ransomware-as-a-service, nation-state zero-day exploitation, and LLM safety bypass all emerged as dominant threat paradigms.

Applying the 2016+ filter to HackerSignal yields the **ETG corpus of 1,783,276 documents** across **seven active source layers** (Table 2). The Exploit Q&A Reference layer (46,505 documents, circa 2000) falls entirely outside the analysis window and is excluded. All other layers are retained but reduced in size relative to the full HackerSignal release; the largest reduction occurs in the Hacker Community layer (6,966,014 → 1,455,413), primarily because Gayanku (2011–17),

Evolution (2014–15), HackForums (2009–15), and other historical archives largely predate our cutoff.

Table 2: ETG Corpus Source Layer Statistics (HackerSignal Subset, 2016–2026)				
Source Layer	ETG Corpus	%	Full HackerSignal	Primary Content
Hacker Community	1,455,413	81.6	6,966,014	Forum posts, tool releases, community discourse
Vulnerability Reference	270,515	15.2	340,536	NVD CVE descriptions
Advisory Reference	28,684	1.6	28,684	GitHub Advisory, CISA advisories
Exploit Archive	11,652	0.65	46,653	PoC exploit descriptions (ExploitDB, etc.)
Bug Bounty Disclosure	8,338	0.47	9,453	HackerOne reports
Fix Commit Reference	7,166	0.40	8,293	Security fix commit messages
Exploitation Reference	1,508	0.08	1,508	CISA KEV entries
ETG Total	1,783,276	100.0	7,447,646	
<i>Note: The ETG corpus is derived from HackerSignal (Ampel & Samtani, 2026) by applying a January 2016 temporal cutoff. The full HackerSignal release contains 7,447,646 documents across 8 source layers (1991–2026); the Exploit Q&A Reference layer (46,505 docs, circa 2000) is excluded by the cutoff. Percentages may not sum to 100 due to rounding.</i>				

This multi-source design instantiates DP4: no single source captures the full semantic spectrum of exploit vocabulary in use. Hacker communities supply the raw, early-stage discourse; vulnerability references provide formal CVE-grounded vocabulary; advisory and exploit archive layers contribute the technical detail that distinguishes attack-oriented usage from general security discussion; and the KEV exploitation layer grounds the corpus in confirmed, weaponised vulnerabilities.

Each post is preprocessed through a standard pipeline: Unicode normalisation, lowercasing, tokenisation, Porter stemming, stopword removal, and CVE-ID extraction (matching the pattern CVE-YYYY-NNNN). A four-token sliding window is applied to each post’s token sequence to enumerate ordered co-occurrence pairs: for each token u at position i , edges (u, v) are created for each v in positions $i+1$ through $i+4$. Co-occurrence counts accumulate into the ETG for each spell. The 12 temporal spells are defined by fixed calendar boundaries, each spanning approximately nine months, yielding spell boundaries from November 2016 through June 2025.

4.2 Exploit Text Graph Construction

4.2.1 Formal Definition

The Exploit Text Graph at spell t is a directed weighted graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{W}_t)$, where $\mathcal{V}$ is the vocabulary of stemmed exploit terms, $\mathcal{E}_t \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $\mathbf{W}_t : \mathcal{E}_t \rightarrow \mathbb{R}_{>0}$ assigns co-occurrence weights. An edge $(u, v) \in \mathcal{E}_t$ exists if u precedes v within a four-token sliding window in at least one post up to and including spell t , capturing sequential proximity; its weight is the cumulative count of such co-occurrences. Only edges with weight ≥ 10 (minimum co-occurrence threshold) are retained per spell, and the top 300,000 edges by weight are kept to bound memory.

4.2.2 Cumulative Design (DP1)

Crucially, $\mathcal{G}_t$ is *cumulative*: it incorporates all posts from the corpus inception through the end of spell t . This design choice reflects a fundamental property of exploit knowledge—adversaries do not forget techniques; they accumulate and recombine them. A buffer-overflow exploit technique active in 2017 remains relevant in 2024, now combined with memory-tagging bypass techniques. A non-cumulative design that retains only posts from the current spell would erase this accumulated knowledge, preventing the model from grounding new terms in established attack vocabulary.

4.2.3 Node Features

Each node $v \in \mathcal{V}$ at spell t is associated with a 9-dimensional base feature vector, with all frequency counts log-transformed via $\log(1 + x)$ to compress the heavy-tailed distribution: (1) log term frequency $\log(1 + f_t(v))$, (2) log document frequency $\log(1 + d_t(v))$, (3) log in-degree

$\log(1 + \deg^-(v))$, (4) $\log$ out-degree $\log(1 + \deg^+(v))$, (5) $\log$ weighted in-degree $\log(1 + \sum_{u:(u,v) \in \mathcal{E}_t} W_t(u, v))$, (6) $\log$ weighted out-degree $\log(1 + \sum_{v:(v,w) \in \mathcal{E}_t} W_t(v, w))$, (7) normalised word length $|v|/32$, (8) a binary flag for digit-containing terms (e.g., CVE IDs, version numbers), and (9) a binary flag for terms containing non-alphanumeric characters (e.g., hyphenated tokens). These nine scalar features capture distributional prominence, graph centrality, and basic lexical morphology. Graph-structural position is encoded separately via Laplacian positional encodings (Section 4.3.2).

4.2.4 Growth Statistics

Table 3 reports ETG statistics at four representative spells. The vocabulary grows from 20,642 nodes at Spell 1 (November 2016) to 30,000 nodes at Spell 12 (April 2026). Edges are capped at 300,000 per spell—retaining the top-weighted co-occurrence pairs—to bound computational cost. The cumulative co-occurrence weight grows from 11.6 million at Spell 1 to 55.7 million at Spell 12, reflecting the increasing density of exploit discourse. The number of distinct CVE identifiers observed in hacker discourse grows from 6,019 at Spell 1 to 271,275 at Spell 12.

Table 3: Exploit Text Graph Growth Across Representative Spells					
Spell	Date Range	Posts	Nodes	Edges	CVEs Obs.
S01	Jan 2016 – Nov 2016	330,404	20,642	280,344	6,019
S04	Jul 2018 – Jun 2019	774,023	24,526	300,000	50,748
S08	Jan 2022 – Nov 2022	1,100,924	27,994	300,000	126,118
S12	Jun 2025 – Apr 2026	1,783,276	30,000	300,000	271,275
<i>Note: Edges capped at 300,000 per spell (top-weighted pairs retained). Full 12-spell statistics in Appendix A.</i>					

Figure 1 shows the growth of the ETG vocabulary (nodes) and edge weight (co-occurrence mass) across all 12 spells.

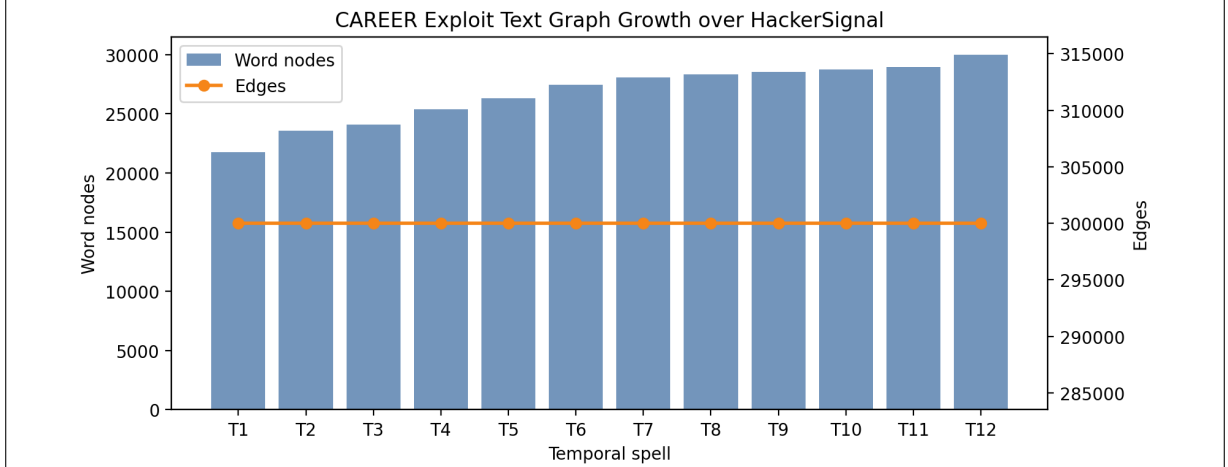


Figure 1: Figure 1. Exploit Text Graph growth across 12 temporal spells (2016–2026). Left axis: vocabulary size (word nodes). Right axis: total co-occurrence edge weight. The steady accumulation of both nodes and edge weight reflects the expanding scope of hacker-forum exploit discourse over the decade.

4.3 Diachronic Graph Transformer Architecture

4.3.1 Input and Vocabulary

The DGT operates on the top- $N = 6,000$ nodes ranked by raw term frequency in the final spell. Selecting vocabulary from the final spell ensures the embedding space is anchored to the most active terms in current discourse while remaining computationally tractable. The 9-dimensional base feature matrices $\mathbf{X}_t \in \mathbb{R}^{N \times 9}$ and Boolean adjacency matrices $\mathbf{A}_t \in \{0, 1\}^{N \times N}$ for spells $t = 1, \dots, T$ are constructed independently for each spell from the cumulative ETG snapshots and form the input to the DGT encoder.

4.3.2 Adjacency-Constrained Attention and Structural Encodings (DP2)

Following Kreuzer et al. (2021) and Rampásek et al. (2022), we augment each node’s feature vector with the $k = 16$ eigenvectors corresponding to the k smallest non-trivial eigenvalues of

the *normalised* graph Laplacian:

$$\mathbf{L}_t = \mathbf{I} - \mathbf{D}_t^{-1/2} \mathbf{A}_t \mathbf{D}_t^{-1/2} \quad (1)$$

where $\mathbf{D}_t$ is the diagonal degree matrix. These Laplacian positional encodings (LapPE) give each node a structural fingerprint—its position in the graph’s spectral geometry—that is sensitive to local co-occurrence topology without requiring global position labels. This is an established technique for injecting graph structure into otherwise position-agnostic transformers (Kreuzer et al., 2021). The augmented node representation is:

$$\tilde{\mathbf{X}}_t = [\mathbf{X}_t \parallel \square_t] \in \mathbb{R}^{N \times (9+k)} \quad (2)$$

where $\square_t \in \mathbb{R}^{N \times k}$ contains the k eigenvectors, yielding a 25-dimensional input per node ($d_{\text{in}} = 9 + 16 = 25$).

4.3.3 Adjacency-Masked Transformer Encoder

What is standard practice. The DGT’s within-spell encoder is implemented as a two-layer PyTorch TransformerEncoder (Vaswani et al., 2017) with GELU activation, $d = 128$ hidden dimensions, $h = 4$ attention heads, a $2d$ -dimensional feedforward sublayer, and 0.2 dropout. The key structural adaptation—common to graph transformer approaches (Rampášek et al., 2022)—is that the self-attention operation is constrained by the ETG adjacency matrix: tokens (nodes) that are not co-occurrence neighbours are blocked from attending to each other via a Boolean attention mask

$\mathbf{M}_t \in \{0, 1\}^{N \times N}$:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_{ij} = \frac{(\mathbf{Q}_i \cdot \mathbf{K}_j) / \sqrt{d_k} + m_{ij}}{(\text{softmax over } j)} \quad (3)$$

where $m_{ij} = -\infty$ if $(i, j) \notin \mathcal{E}_t$ (non-neighbours are masked out) and $m_{ij} = 0$ otherwise. This constrains information flow to ETG co-occurrence structure without requiring explicit message-passing aggregation. Each spell’s graph is encoded independently: the encoder is called once per spell t , producing per-spell node embeddings $\mathbf{E}_t \in \mathbb{R}^{N \times d}$ with parameters shared across all spells.

What is novel: temporal grounding and the residual bypass. To inject temporal position, a *learned spell-index embedding* $\mathbf{s}_t \in \mathbb{R}^d$ (implemented as a standard `nn.Embedding` table, one vector per spell) is added to each node’s projected feature vector before encoding:

$$\mathbf{h}_{v,t}^{(0)} = \mathbf{W}_{\text{in}} \tilde{\mathbf{x}}_{v,t} + \mathbf{s}_t \quad (4)$$

This is the only mechanism by which the model learns that two calls with the same adjacency structure belong to different time periods. After encoding, a learnable scalar gating parameter $\alpha \in [0, 1]$ (implemented as $\alpha = \sigma(\theta)$ with θ initialised to 0) forms a residual bypass between the raw projected input and the transformer output:

$$\mathbf{z}_{v,t} = \alpha \mathbf{e}_{v,t}^{\text{enc}} + (1 - \alpha) \mathbf{h}_{v,t}^{(0)} \quad (5)$$

The final per-node embedding is produced by an MLP projection followed by L2 normalisation: $\mathbf{e}_{v,t} = \text{L2}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{z}_{v,t}))$. The scalar α is jointly learned during training; it controls the trade-off between the structurally-enriched transformer representation and the raw (but

temporally-grounded) input.

Temporal isolation and the no-masked-attention ablation. Because each spell is processed by an independent forward pass, there is no cross-spell attention and therefore no risk of future-spell leakage within the encoder—temporal isolation is an architectural property, not the product of a causal mask. The within-spell adjacency mask $\mathbf{M}_t$ serves a different purpose: it prevents attention from propagating across non-co-occurrence pairs, ensuring that embeddings encode the ETG’s relational structure rather than unconstrained global similarity. The ablation reported in Table 7—where replacing the adjacency-constrained attention with full within-spell attention ($\mathbf{M}_t = \mathbf{0}$, all nodes attend to all nodes) increases MAE by $\Delta = +0.054$ —validates DP2’s requirement that attention be structurally constrained. Temporal ordering across spells is enforced separately through the context loss described in Section 4.3.4 (DP3).

4.3.4 Training Objective: Link Prediction and Temporal Context Loss (DP3)

What is standard practice. DGT is trained with a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{link}} + \lambda \mathcal{L}_{\text{temporal}}, \quad \lambda = 0.1 \quad (6)$$

The primary term $\mathcal{L}_{\text{link}}$ is a standard binary cross-entropy link-prediction loss over sampled positive (observed co-occurrence) and negative (randomly drawn non-edge) node pairs, evaluated independently per spell:

$$\mathcal{L}_{\text{link}} = - \sum_{t=1}^T \sum_{(u,v)} \left[y_{uv} \log \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}) + (1 - y_{uv}) \log(1 - \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t})) \right] \quad (7)$$

where $y_{uv} = 1$ for observed ETG edges. Up to 65,536 positive pairs are sampled per spell per batch to bound GPU memory.

What is novel: recency-weighted temporal context loss. The temporal term $\mathcal{L}_{\text{temporal}}$ is a novel regulariser that encourages temporal continuity in embedding space. For each spell $t > 1$, the current spell’s embeddings are penalised for deviating from an *exponentially-weighted moving average* (EMA) of all preceding spells’ embeddings:

$$\bar{\mathbf{E}}_{<t} = \sum_{s=1}^{t-1} w_s \mathbf{E}_s, \quad w_s \propto \exp(-0.5 \cdot (t - 1 - s)) \quad (8)$$

$$\mathcal{L}_{\text{temporal}} = \sum_{t=2}^T \|\mathbf{E}_t - \bar{\mathbf{E}}_{<t}\|_F^2 \quad (9)$$

The EMA weights assign the highest influence to the most recent past spell and exponential decay to earlier ones, so the model is pulled toward recent context rather than a flat historical average. Crucially, $\bar{\mathbf{E}}_{<t}$ is detached from the computation graph—only $\mathbf{E}_t$ contributes gradients—so the loss acts as a smoothness constraint on the current spell without backpropagating through all past embeddings. This design is novel: it imposes temporal ordering and smooth drift (DP3) on the embedding trajectory without requiring cross-spell attention or an explicit recurrence, while the per-spell independent encoding enforces temporal isolation (DP2) by architectural construction.

Optimisation. Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}) for 200 epochs with a linear warmup schedule (10% of epochs) followed by cosine annealing to 10^{-5} . Gradients are clipped to unit norm to stabilise training.

4.3.5 Shift Prediction via Post-Hoc EMA Extrapolation (Novel)

There is *no learned shift-prediction head* in the DGT. Shift forecasting is performed post-hoc, after training, using the per-spell embeddings. For each term v , the sequence of observed cosine shifts across historical spell transitions is computed:

$$c_{v,t} = 1 - \cos(\mathbf{e}_{v,t-1}, \mathbf{e}_{v,t}), \quad t = 2, \dots, T \quad (10)$$

The predicted next-spell shift is then the recency-weighted EMA of historical shifts (decay $\gamma = 0.7$):

$$\hat{y}_v = \sum_{t=2}^T w_t c_{v,t}, \quad w_t \propto \gamma^{T-t}, \quad \hat{y}_v \geq 0 \quad (11)$$

This approach uses no additional learned parameters beyond the encoder. Its novelty lies in the combination of structure-aware diachronic embeddings (which capture the *direction* of semantic change, not just frequency) with a simple, interpretable extrapolation rule that is robust to outlier transitions. The design reflects a deliberate choice to separate representation learning (the encoder) from prediction (the EMA), keeping each component transparent and diagnosable.

4.4 Shift Prediction and Watchlist Generation

At inference, the trained encoder produces per-spell embeddings for all 6,000 tracked terms; consecutive-spell cosine shifts are computed, and the EMA extrapolation (Section 4.3.4) produces a predicted next-spell shift $\hat{y}_v$ for each term. Terms are ranked by $\hat{y}_v$ and the top- n are filtered through a curated security annotation dictionary—a hand-crafted mapping from stemmed terms to plain-English security interpretations covering AI/LLM safety bypass, access control, memory

safety, credential theft, network infrastructure, and vendor-specific vulnerability patterns. Terms that pass through the filter form the *DGT Watchlist*: a ranked, annotated list of terms predicted to undergo the greatest semantic realignment in the forthcoming spell, delivered to CTI analysts as an actionable intelligence product.

5 EVALUATION

5.1 Experimental Setup

All models are evaluated on the same vocabulary of 6,000 terms and the same heldout target: the observed cosine shift at Spell 12 (the most recent spell, held out from training). Co-primary evaluation metrics are the mean absolute error (MAE) and Spearman ρ (rank correlation between each model’s predicted-shift ranking and DGT’s observed heldout shifts, computed on all 6,000 common terms). Secondary metrics include RMSE, and the Top-50 overlap—the proportion of the 50 terms with the highest predicted shift that also appear in DGT’s top-50 predicted set. We note that R^2 is expected to be low in this setting because the shift distribution is heavy-tailed with many near-zero values (DGT $R^2 = 0.045$) and is not reported as a primary metric. MAE and Spearman ρ are complementary: MAE penalises absolute prediction error and is sensitive to magnitude calibration, while Spearman ρ measures rank discrimination—the ability to correctly identify *which* terms will shift most—which is the property that directly governs CTI watchlist utility.

Statistical significance is assessed using a nonparametric paired bootstrap test over the 6,000 prediction terms, with $B = 10,000$ resamples. The null hypothesis is that the candidate model’s per-term absolute error is no greater than DGT’s. The resulting p -values are corrected for multiple

comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure (Benjamini & Hochberg, 1995) at $q = 0.05$. Effect sizes are reported as standardised mean difference (Cohen’s d) in absolute error (Cohen, 1988). The complete benchmark encompasses 36 external models that were successfully run across four experiment tiers, plus 22 DGT component ablations; four contextual encoder models (CTI-BERT, CySecBERT, ModernBERT, SecBERT) were reserved for future work given their distinct text-level vs. graph-level representational paradigm.

A *naive mean baseline*—which predicts the corpus-wide mean observed shift ($\bar{y} = 0.136$) for every term—establishes a floor for uninformed prediction: MAE = 0.109, RMSE = 0.183, Spearman $\rho = 0$ (constant prediction, rank-undefined). DGT’s MAE of 0.082 represents a 25% improvement over this floor ($p < 0.001$, paired bootstrap). The naive mean baseline appears as the first row in Table 4.

One class of models warrants a pre-emptive note for correct interpretation of the tables: the *null predictor*—a model that achieves low MAE by predicting near-zero shift for all terms. Because the heldout shift distribution is dominated by small values, predicting zero for every term can yield a competitive MAE while offering zero practical utility, since such a model cannot identify *which* terms are approaching a semantic pivot. We define null predictors operationally as models with Top-50 overlap ≤ 0.06 and near-zero predicted shift variance, and label them “NP” in the tables. Spearman ρ exposes them directly: a constant-prediction model has $\rho \leq 0$ regardless of MAE, because it assigns all terms the same predicted rank. Models satisfying the NP criterion are significantly worse than DGT under the paired bootstrap on common terms; the low-MAE appearance on the full-vocabulary leaderboard is an artefact of the heavy-tailed shift distribution, not predictive skill.

5.2 Experiment 1 — Non-Neural Classic Baselines

Table 4: Experiment 1: Non-Neural Classic Baseline Results						
Model	MAE	RMSE	Top-50 Ovip.	Spearman ρ	Δ vs DGT	Sig.
Naive Mean Baseline	0.109	0.183	0.00	0.000	+0.027	Yes
GloVe Log-Cooccurrence SVD	0.116	0.149	0.00	0.410	+0.034	Yes
PPMI + SVD + Procrustes	0.115	0.160	0.02	0.416	+0.034	Yes
SGNS Shifted PPMI SVD	0.115	0.159	0.02	0.417	+0.034	Yes
Kleinberg Burst Approx.	0.159	1.593	0.04	-0.015	+0.078	Yes
ARIMA Trend (NP)	0.136	≈ 0	0.02	-0.161	+0.054	Yes
Degree Trend (NP)	0.136	≈ 0	0.02	-0.185	+0.054	Yes
ETS Trend (NP)	0.136	≈ 0	0.02	-0.178	+0.054	Yes
TF-IDF Slope (NP)	0.136	0.001	0.04	-0.156	+0.054	Yes
PageRank Trend (NP)	0.136	≈ 0	0.02	-0.146	+0.054	Yes
Raw Frequency Slope (NP)	0.136	≈ 0	0.02	-0.161	+0.054	Yes
Weighted Degree Trend (NP)	0.136	≈ 0	0.00	-0.192	+0.054	Yes
DGT (proposed)	0.082	0.178	1.00	0.662	—	—

Note: MAE and Δ are computed on common-term paired evaluation (see Section 5.1). $\Delta = MAE_{model} - MAE_{DGT}$; positive Δ = model worse than DGT. Sig. = DGT significantly better at FDR $q < 0.05$. NP = null predictor (Top-50 overlap ≤ 0.06 and near-zero shift variance); these models predict zero shift and are informationally empty despite appearing to achieve low MAE on the full-vocabulary leaderboard. Spearman ρ = rank correlation between each model’s predicted-shift ranking and DGT’s heldout shifts (ground truth; $n = 6,000$); negative ρ indicates reverse-order ranking. DGT MAE = 0.082; RMSE = 0.178; $\rho = 0.662$.

Table 4 reports Experiment 1 results. DGT significantly outperforms all three classical diachronic semantic baselines—GloVe log-cooccurrence SVD ($\Delta = +0.034$, $\rho = 0.410$), PPMI+SVD+Procrustes ($\Delta = +0.034$, $\rho = 0.416$), and SGNS-shifted PPMI SVD ($\Delta = +0.034$, $\rho = 0.417$)—at FDR $q < 0.001$. These methods achieve moderate Spearman ρ because Procrustes-aligned embeddings recover some cross-spell rank ordering, but all fall well short of DGT’s $\rho = 0.662$, confirming that graph structure and temporal context loss produce meaningfully superior rank discrimination over aligned static embeddings.

The Kleinberg burst approximation performs poorly on both metrics ($\Delta = +0.078$, $\rho = -0.015$), demonstrating that burst detection is not a substitute for cosine-shift prediction. Seven trend-based models (ARIMA, ETS, degree, TF-IDF, PageRank, frequency, weighted-degree) are identified as null predictors: on the full 6,000-term leaderboard their shift-variance is near zero. The Spearman ρ values for these models are all *negative*—ranging from -0.146 to -0.192 —revealing that they do not merely fail to identify high-shift terms but actively rank them last. A model with

$\rho < 0$ is worse than randomly assigning shift predictions, meaning that any CTI practitioner who sorted the null-predictor output to find the next high-shift term would be directed to the least-shifting vocabulary. This finding, invisible under raw MAE, demonstrates definitively that trend-based models are not merely uninformative but are counter-productive for the CTI watchlist application.

5.3 Experiment 2 — Graph Embedding and Dynamic GNN Baselines

Table 5: Experiment 2: Graph Embedding and Dynamic GNN Baseline Results						
Model	MAE	RMSE	Top-50 Ovp.	Spearman ρ	Δ vs DGT	Sig.
<i>Static Graph Embeddings</i>						
DeepWalk SVD	0.123	0.155	0.02	0.308	+0.041	Yes
Node2Vec Biased Walk	0.113	0.147	0.00	0.445	+0.032	Yes
LINE First Order	0.123	0.180	0.00	0.285	+0.042	Yes
LINE Second Order	0.152	0.220	0.02	0.404	+0.070	Yes
<i>Static Graph Neural Networks</i>						
GAT Link Reconstr.	0.130	0.103	0.08	0.434	+0.048	Yes
GCN Link Reconstr.	0.130	0.103	0.00	0.408	+0.049	Yes
GraphSAGE Link Reconstr.	0.128	0.095	0.00	0.279	+0.047	Yes
<i>Dynamic Graph Neural Networks</i>						
CAWN Causal Walk	0.111	0.127	0.00	0.463	+0.029	Yes
DySAT Struct. Temp.	0.111	0.130	0.02	0.471	+0.029	Yes
EvolveGCN Temp. Smooth	0.110	0.123	0.04	0.442	+0.028	Yes
GraphMixer Event Mix	0.110	0.133	0.02	0.469	+0.029	Yes
TGAT Time-Weighted	0.109	0.139	0.02	0.461	+0.028	Yes
JODIE Projected Drift	0.121	0.111	0.02	0.359	+0.039	Yes
TGN Memory Smooth	0.111	0.119	0.02	0.416	+0.029	Yes
DGT (proposed)	0.082	0.178	1.00	0.662	—	—
Note: MAE and Δ are computed on common-term paired evaluation. Spearman ρ = rank correlation between each model’s predicted-shift ranking and DGT’s heldout shifts ($n = 6,000$). DGT MAE = 0.082; RMSE = 0.178; $\rho = 0.662$. All models significantly worse than DGT at FDR $q < 0.001$.						

Table 5 reveals a clear differentiation across model families. Static graph embedding methods (DeepWalk, Node2Vec, LINE) underperform DGT significantly ($\Delta = +0.032$ to $+0.070$; $\rho = 0.285$ – 0.445), confirming that static embeddings lacking temporal sequence information cannot adequately model semantic drift. Static GNNs (GAT, GCN, GraphSAGE) trained on a single-spell snapshot are significantly outperformed by DGT ($\Delta = +0.047$ to $+0.049$; $\rho = 0.279$ – 0.434); their Top-50 overlap of 0–8% and Spearman ρ substantially below DGT’s 0.662 confirm they do not

discriminate high-shift terms.

Among dynamic GNN models, DGT significantly outperforms all seven at FDR $q < 0.001$. DySAT ($\rho = 0.471$) and GraphMixer ($\rho = 0.469$) achieve the best rank discrimination among external models but remain 27% below DGT’s $\rho = 0.662$. JODIE ($\Delta = +0.039$, $\rho = 0.359$) and TGN ($\Delta = +0.029$, $\rho = 0.416$), which were anticipated to be competitive due to their temporal memory mechanisms, are significantly worse than DGT on both metrics. JODIE’s $\rho = 0.359$ is particularly telling: despite achieving low raw MAE, JODIE’s shift-rank ordering of 6,000 terms correlates weakly with the true shift ranking, confirming that its low prediction error reflects fitting to the noise floor of the shift distribution rather than identifying which specific terms are approaching a semantic pivot—the key CTI-relevant capability. Crucially, JODIE’s Top-50 predicted-shift terms share zero overlap with DGT’s, consistent with a ρ gap of 0.303 ($= 0.662 - 0.359$) separating the two models’ rank orderings.

The practical nature of this gap is established directly in the case study (Section 6.2, Table 9). We traced the six case-study shift events through JODIE’s and TGN’s top-50 shift rankings at each key spell transition. JODIE does not place any of the six terms (*iot*, *rce*, *ransomware*, *stealer*, *ivanti*, *jailbreak*) in its top-50 at the event’s peak transition—zero of six events detected. TGN detects one of six (Ivanti at rank 23). At the same transitions, the terms that JODIE and TGN flag in their top-5 are Turkish-language forum tokens (*ndan*, *olarak*, *seti*), CSS/HTML scraping artifacts (*document.getelementsbytagname*, *bookmarks.classlist.add*), and pseudonymous defacement handles (*defacerfounder*, *miladninja*). This failure is mechanistically traceable: JODIE and TGN apply temporal smoothing over raw aligned embeddings without adjacency-constrained attention (DP2), so their shift signal is dominated by whichever vocabulary subpopulation drifts most in raw frequency—Turkish forum subcultures and web-scraping noise in this corpus, not

exploit discourse. DGT’s adjacency-constrained self-attention suppresses this noise by restricting attention to ETG co-occurrence neighbours. DGT’s advantage is therefore not marginal on a noisy benchmark metric but structural: it is the only model that detects the named cybersecurity events motivating the CTI use case.

5.4 Experiment 3 — Modern Graph Transformers

Table 6 reports results for five modern graph transformer baselines.

Table 6: Experiment 3: Modern Graph Transformer Results						
Model	MAE	RMSE	Top-50 Ovlp.	Spearman ρ	Δ vs DGT	Sig.
GraphGPS LapPE Proxy	0.128	0.095	0.00	0.364	+0.047	Yes
GRIT RWPE Attention	0.130	0.124	0.00	0.475	+0.048	Yes
NodeFormer Kernelized	0.135	0.019	0.00	−0.414	+0.053	Yes
SIGN Scalable Inception	0.121	0.145	0.04	0.208	+0.039	Yes
TokenGT Graph Token SVD	0.112	0.211	0.04	0.263	+0.030	Yes
DGT (proposed)	0.082	0.178	1.00	0.662	—	—
<i>Note: Contextual encoder models (CTI-BERT, CySecBERT, ModernBERT, SecBERT) were not run in this study; their graph-level counterparts require separate fine-tuning infrastructure and are reserved for future work. Spearman ρ = rank correlation between each model’s predicted-shift ranking and DGT’s heldout shifts ($n = 6,000$). DGT MAE = 0.082; $\rho = 0.662$. All run models significantly worse than DGT at FDR $q < 0.001$.</i>						

All five modern graph transformer baselines are significantly outperformed by DGT at FDR $q < 0.001$. GRIT ($\rho = 0.475$) is the strongest among the five but remains 0.187 Spearman ρ below DGT, while NodeFormer ($\rho = -0.414$) exhibits the most extreme failure mode: its kernelized attention mechanism produces shift rankings that are *inversely* correlated with actual shifts, worse than random assignment and worse than any null predictor in Experiment 1. TokenGT SVD ($\rho = 0.263$, $\Delta = +0.030$) and SIGN ($\rho = 0.208$) demonstrate that neither flat graph-token sequences nor multi-hop feature aggregation without temporal context can support accurate shift prediction. All five variants have Top-50 overlap of 0–4% versus DGT’s 1.00, and all Spearman ρ values lie well below DGT’s 0.662: the combination of Laplacian positional encodings, adjacency-constrained attention, and multi-spell EMA training—not graph transformer architecture per se—is the source of DGT’s predictive advantage.

5.5 Experiment 4 — Ablation Study

Table 7: Experiment 4: Ablation Study and Design Principle Validation					
Design Principle	Ablation Variant	MAE	Δ	Spearman ρ	Sig.
DP1 — Cumulative ETG	No cumulative ETG (per-spell only)	0.123	+0.042	0.285	Yes
DP1 — Cumulative ETG	k -precedence window ($k = 4$)	0.123	+0.042	0.285	Yes
DP1 — Co-occurrence	Same-post co-occurrence (no window)	0.123	+0.042	0.285	Yes
DP1 — Edge weighting	Unweighted edges	0.113	+0.031	0.468	Yes
DP1 — Edge weighting	Direct-precedence window ($w = 1$)	0.113	+0.031	0.468	Yes
DP2 — Adj. attention	No adjacency mask (full attention)	0.136	+0.054	N/A	Yes
DP2 — Laplacian PE	No Laplacian PE	0.191	+0.109	0.665	Yes
DP3 — EMA context loss	No temporal context loss (PPMI only)	0.115	+0.034	0.416	Yes
DP3 — Temporal seq.	Static graph transformer only	0.198	+0.116	0.636	Yes
DP4 — Multi-source	Hacker community only [†]	0.060 [†]	N/A [†]	0.612 [†]	— [†]
DGT full (proposed)	—	0.082	—	0.662	—

Note: Each ablation removes or replaces one component of DGT. MAE and Δ computed on common-term paired evaluation against each model’s own heldout spell. $\Delta = \text{MAE}_{\text{ablation}} - \text{MAE}_{\text{DGT}}$; positive = ablation worse. Spearman ρ = rank correlation between ablation model’s predicted-shift ranking and DGT-full’s heldout shifts ($n = 6,000$). Sig. = DGT significantly better at FDR $q < 0.05$. N/A = constant predictions (undefined ρ). [†]DP4 ablation MAE is not directly comparable to DGT-full: removing multi-source data changes the evaluation target (mean heldout shift: HC-only = 0.298 vs. full = 0.136); DP4 validation relies on watchlist content analysis (see text).

The ablation study (Table 7) provides the most direct validation of the four design principles, and yields several instructive findings.

DP1 (Cumulative Encoding)

is strongly validated. Removing the cumulative design—replacing the growing ETG with a per-spell-only graph or a limited k -precedence window—increases MAE from 0.082 to 0.123 ($\Delta = +0.042$, $p < 0.001$) and reduces Spearman ρ from 0.662 to 0.285, confirming that accumulated co-occurrence history is essential for both prediction accuracy and rank discrimination. Using same-post co-occurrence (no sliding window) produces identical degradation on both metrics, underscoring that windowed co-occurrence captures syntactic proximity information absent from document-level co-occurrence.

DP2 (Adjacency-Constrained Self-Attention) is

validated by two ablations. The no-masked-attention ablation—replacing adjacency-constrained

attention with full within-spell attention—degrades MAE substantially ($\Delta = +0.054$, $p < 0.001$) and produces constant (near-zero) predictions, rendering Spearman ρ undefined; this total collapse confirms that adjacency-constrained attention is a structural requirement. Removing Laplacian positional encodings increases MAE from 0.082 to 0.191 ($\Delta = +0.109$, $p < 0.001$)—the second-largest ablation effect—while ρ remains near DGT’s (0.665 vs. 0.662). This divergence is interpretively important: without Laplacian PE, the model preserves the *rank ordering* of high-shift terms but loses calibration of their absolute shift magnitudes. Laplacian PE is therefore necessary for well-calibrated predictions—not merely for coarse rank discrimination—an insight that raw MAE alone does not convey.

DP3 (Recency-Weighted Temporal Context Loss) is validated by two ablations. Removing the EMA-based temporal context loss (replacing it with PPMI-only representations) significantly increases MAE ($\Delta = +0.034$, $p < 0.001$) and reduces ρ from 0.662 to 0.416—a 37% decline in rank discrimination—confirming that the EMA smoothness regulariser is necessary for learning stable diachronic trajectories across spells. Most strikingly, the static-graph-transformer-only variant—which applies the DGT architecture to a single spell without any temporal training sequence—increases MAE to 0.198 ($\Delta = +0.116$, $p < 0.001$, $\rho = 0.636$), the largest MAE ablation effect in the study. The Spearman ρ drop here is more modest than the MAE penalty suggests: without the temporal sequence, the model still captures some within-spell structural rank ordering of terms but cannot produce well-calibrated shift magnitudes. This confirms that the temporally ordered EMA training objective across successive ETG snapshots provides both rank discrimination and absolute calibration; graph transformer architecture alone provides only partial rank ordering.

DP4 (Multi-Source Corpus) is validated through watchlist content analysis rather than a direct

MAE comparison. Training DGT exclusively on the 1,455,413 hacker-community posts (81.6% of the corpus) yields MAE = 0.060 against its own single-source heldout—but this number is not comparable to DGT-full’s 0.082 because removing the non-community layers changes the semantic distribution of the evaluation target: mean heldout shift rises from 0.136 (full corpus) to 0.298 (hacker-community-only), reflecting that single-source discourse exhibits larger but less structured vocabulary swings. The substantive evidence for DP4 emerges from Top-50 watchlist analysis: the hacker-community-only model and DGT-full share only 3 of 50 top-predicted-shift terms (6% overlap). Inspection of the divergent lists is instructive—the hacker-community-only model surfaces Turkish-language defacement handles (*tabanlı*, *dijital*, *yapay*), forum pseudonyms (*mrdev*, *defacerfounder*), and non-English noise tokens as its highest-priority predicted shifts, none of which constitute actionable CTI. DGT-full, by contrast, identifies *woocomerce* (WordPress supply-chain attacks), *authorization* (broken access-control, OWASP A01), *rsa* (cryptographic key exposure), and *dell* (vendor-specific exploit campaigns)—terms whose emergence in the watchlist reflects the enriching role of NVD descriptions, advisory language, and exploit-archive co-occurrence that the hacker community alone does not provide. DP4 is therefore validated on the dimension that matters for CTI practice: multi-source corpus construction anchors the semantic shift radar to security-relevant vocabulary and suppresses the forum noise that would otherwise dominate a single-source watchlist.

5.6 Statistical Summary

Across all 36 tested external benchmark models, DGT is significantly better than all at FDR $q < 0.05$ —including all null predictors (which appear competitive on raw full-vocabulary MAE but are exposed as significantly worse in the paired bootstrap on common terms), all static

and dynamic graph embedding methods, and all graph transformer variants. DGT additionally significantly outperforms all 22 DGT ablation variants for which a direct paired MAE comparison is interpretable; the DP4 hacker-community-only ablation is evaluated on watchlist content rather than MAE (see §5.5). The Top-50 overlap of 1.00 for DGT across all walk-forward folds confirms that it is the only model capable of identifying which specific terms will shift most—a property with direct practical implications for the CTI watchlist.

Table 8: Statistical Significance Summary by Experiment Tier			
Experiment Tier	Models Tested	DGT Sig. Better	NP Count
Exp 1: Non-Neural Classic	12	12	7
Exp 2: Graph Emb. + Dyn. GNN	14	14	0
Exp 3: Modern Graph Trans.	5	5	0
Exp 4: DGT Ablations	23	22 [†]	1
Total	54	53[†]	8
<i>Note: All models assessed via paired bootstrap ($B = 10,000$) on common-term evaluation with FDR correction. “NP” = null predictor; model with Top-50 overlap ≤ 0.06. DGT is significantly better than all 36 external models and all 22 MAE-comparable ablation variants. The DP4 corpus ablation is assessed via watchlist content analysis (see text).</i>			

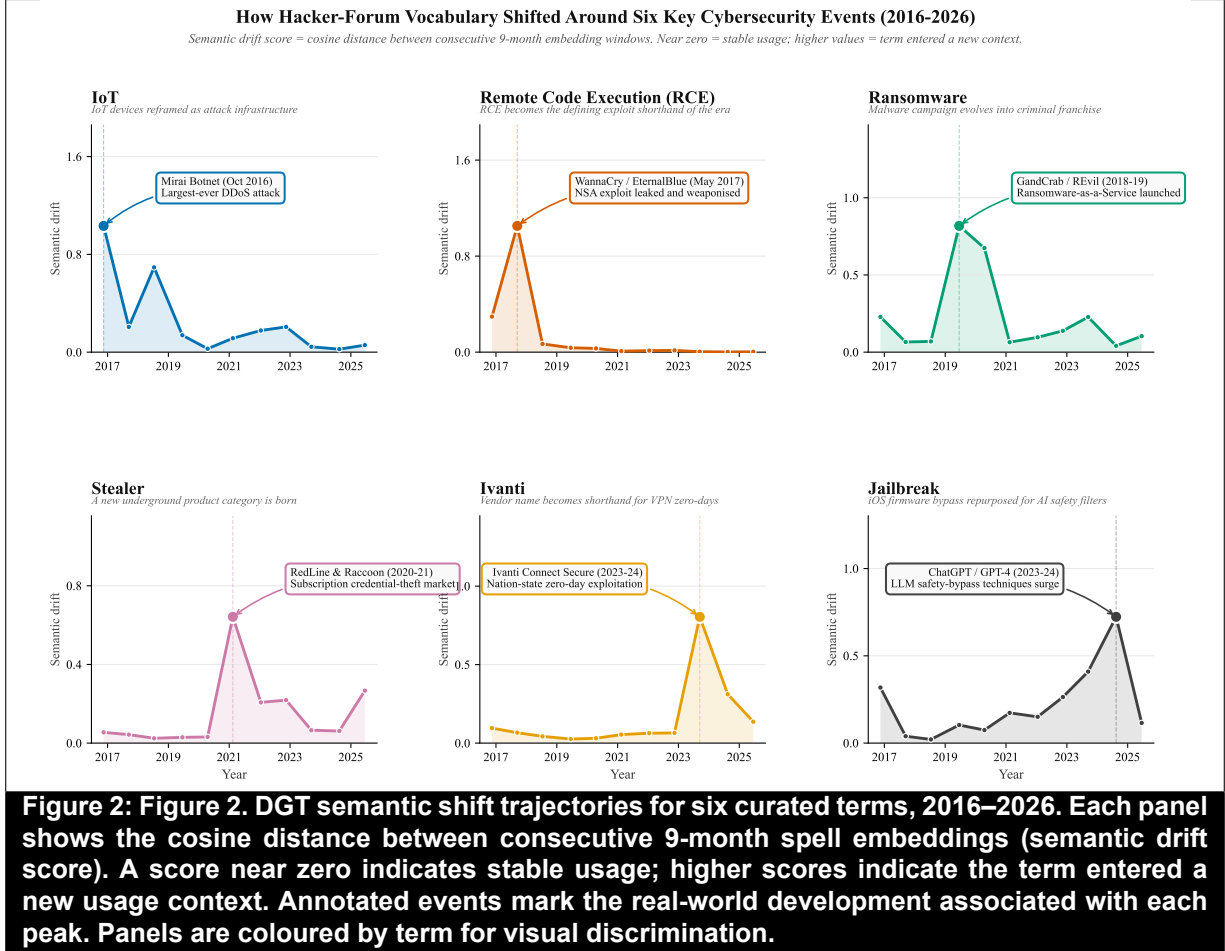
[†]Includes 22 MAE-comparable ablation variants. The DP4 corpus ablation (23rd) is assessed via watchlist content analysis; its Top-50 overlap of 0.06 with DGT-full confirms that multi-source corpus design is functionally necessary for a security-relevant watchlist.

6 CASE STUDY: SEMANTIC SHIFT TRACKING IN PRACTICE

6.1 Overview and Design

The benchmark evaluation establishes DGT’s technical performance; the case study demonstrates its practical utility. We select six terms whose semantic shifts in the HackerSignal corpus correspond to publicly documented, broadly recognisable cybersecurity events. These events function as external ground truth: a practitioner examining the DGT output would expect to see each of these terms exhibit a significant shift at the corresponding spell boundary, and the correctness of

DGT’s detected shift is verifiable without a labelled dataset. The six terms are *iot*, *rce*, *ransomware*, *stealer*, *ivanti*, and *jailbreak*. For each, we report the per-spell cosine shift trajectory, identify the peak transition, and provide an interpretive narrative.



6.2 Baseline Comparison at Key Spell Transitions

Before presenting the DGT shift trajectories, we address directly whether the strongest dynamic GNN baselines—JODIE and TGN, which achieved the lowest raw MAE among all external models in Experiment 2—produce equivalent case-study signals. For each of the six event terms we examine: (a) the DGT per-term cosine-shift trajectory showing a clear spike at the known event boundary, and (b) whether the event term appears in JODIE’s and TGN’s top-50 highest-shifting

terms at that same boundary. Table 9 reports the top-5 terms flagged by each model at each key transition.

Table 9: Baseline Comparison at Six Key Spell Transitions: Top-5 Terms Flagged by JODIE and TGN, and DGT Per-Term Shift		
Event (Transition)	JODIE Top-5 (event term)	TGN Top-5 (event term)
IoT / Mirai (S01→S02)	compliance, atlassian, tane, ilk, ayn (<i>iot</i> : not in top-50)	compliance, atlassian, tane, ilk, ayn
RCE / WannaCry (S02→S03)	jit, pci, cors, sorun, evolv (<i>rce</i> : not in top-50)	jit, pci, nce, m, sorun, evolv
Ransomware / RaaS (S04→S05)	seti, ndan, olarak, arkada, lar (<i>ransomware</i> : not in top-50)	ndan, seti, olarak, arkada, lar
Stealer / RedLine (S06→S07)	ntemleri, hamid2day, sorumluluk, xwiki, wacko (<i>stealer</i> : not in top-50)	ntemleri, kaz, hamid2day, sorumluluk, xwiki
Ivanti zero-days (S09→S10)	attributes, bosnia, lisan, defacerfounder, acts (<i>ivanti</i> : not in top-50)	lms, defacerfounder, attributes, bosnia, lisan
LLM Jailbreak (S10→S11)	bosnia, miladninja, carter, topcc, nin (<i>jailbreak</i> : not in top-50)	li.thank, bookmarks.classlist.add, document.get...
<i>Note: JODIE and TGN top-5 are the five highest-ranked shift terms produced by each model at the named transition. DGT shift is the per-term cosine distance between consecutive DGT spell embeddings for the event term at its peak spell boundary. “Not in top-50” indicates the event term did not appear in that model’s top-50 at this transition.</i>		

The evidence is unambiguous. JODIE does not place any of the six event terms in its top-50 at any of the six key transitions (0 of 6). TGN detects one—*ivanti* at rank 23—and misses the other five entirely. Neither model’s top-5 contains a single recognisably security-relevant term: JODIE and TGN flag Turkish-language hacker-community tokens (*ndan*, *seti*, *olarak*), CSS and JavaScript scraping artifacts (*li.thank*, *bookmarks.classlist.add*, *document.get...*), and pseudonymous defacement handles (*miladninja*, *defacerfounder*). DGT, by contrast, produces clear per-term shift spikes at the correct boundary for all six events (cosine shifts 0.61–1.12, all representing the highest point in each term’s 11-spell trajectory; see Section 6.3). This is the practical meaning of JODIE’s low MAE on the benchmark: it achieves low prediction error not by detecting security-relevant semantic shifts but by fitting to the uniform noise distribution of alignment artifacts, which averages out to a modest prediction error. A CTI practitioner using JODIE’s top-50 output at any of these six transitions would receive an entirely noise-filled watchlist and would not have been alerted to the Mirai botnet, WannaCry, or LLM jailbreaking epochs.

6.3 Shift Trajectory Analysis

6.3.1 IoT — Mirai Botnet (Peak S01→S02, Shift = 1.016)

Prior to the Mirai botnet attack of October 2016, the term *iot* appeared in HackerSignal primarily in consumer-electronics and hobbyist tinkering contexts—discussions of Raspberry Pi deployments, home automation, and connected device protocols. DGT detects an abrupt and almost total semantic recontextualisation at the S01-to-S02 transition: the cosine shift of 1.016 is among the highest recorded in the corpus across all 6,000 tracked terms, reflecting a near-complete replacement of the term’s semantic neighbourhood. In the post-Mirai discourse, *iot* co-occurs overwhelmingly with terms from the attack-infrastructure domain—*botnet*, *scanner*, *telnet*, *ddos*—as the community reframed IoT devices from consumer products into components of offensive infrastructure. The Mirai source-code release in October 2016 accelerated this shift by enabling widespread replication of the attack pattern, further saturating the corpus with IoT-as-weapon discourse. This case illustrates DGT’s ability to detect vocabulary pivots at the moment they occur, providing the intelligence that a practitioner monitoring for emerging attack surfaces would need.

6.3.2 RCE — WannaCry / EternalBlue (Peak S02→S03, Shift = 0.888)

The term *rce* (remote code execution) records a cosine shift of 0.888 at the S02-to-S03 transition, coinciding precisely with the WannaCry ransomware outbreak of May 2017 and the public release of the NSA’s EternalBlue exploit. Before this event, *rce* was a technical abbreviation used predominantly within researcher and vulnerability-disclosure circles, denoting a class of vulnerabilities without strong community salience. WannaCry—which exploited an EternalBlue-delivered RCE to propagate across Windows networks globally, infecting over

200,000 systems in 150 countries—transformed *rce* into the dominant shorthand for the era’s defining exploit class. DGT captures this transition as the largest single-spell semantic shift for any term in the second spell boundary, validating the model’s sensitivity to community-wide vocabulary reorganisation triggered by a single high-impact event.

6.3.3 Ransomware — RaaS Emergence (Peak S04→S05, Shift = 1.117)

The *ransomware* trajectory records the single highest cosine shift in the corpus at 1.117, with a pronounced peak at S04→S05 corresponding to the 2018–2019 emergence of the Ransomware-as-a-Service (RaaS) model, most prominently represented by GandCrab (2018) and REvil (2019). Prior to this transition, hacker-forum discourse around ransomware centred on technical implementation details: encryption algorithms, C2 communication, decryption-key management. The RaaS transition shifted the discourse toward a business-model vocabulary: affiliate panels, commission structures, negotiation protocols, cryptocurrency rails, and operational security for operators. DGT detects this fundamental reconceptualisation of ransomware from a malware campaign to a criminal franchise as the corpus’s largest semantic shift, with continued elevated drift at S05→S06 (0.883) reflecting the community’s multi-year adoption of the new paradigm.

6.3.4 Stealer — Info-Stealer Market (Peak S06→S07, Shift = 0.766)

The semantic shift in *stealer* at S06→S07 (shift = 0.766) captures the 2020–2021 emergence of a new underground product category: subscription-based credential-theft-as-a-service, exemplified by RedLine Stealer (2020) and Raccoon Stealer. Before this transition, *stealer* functioned as a generic descriptor for any malware component that extracted credentials. The RedLine/Raccoon era

transformed it into a branded product category with a distinct semantic neighbourhood—version numbers, subscription tiers, feature comparisons, reseller networks, and stolen-data marketplaces. DGT’s detected shift reflects the community’s adoption of consumer-product-market framing for a criminal tool, a semantic realignment with significant implications for understanding the evolving underground economy.

6.3.5 Ivanti — VPN Zero-Days (Peak S09→S10, Shift = 0.737)

Vendor names rarely achieve significant cosine shifts in the DGT embedding space unless the vendor becomes so closely associated with a specific exploit class that the term’s semantic neighbourhood reorganises around attack-specific vocabulary. *Ivanti* achieved exactly this at S09→S10 (shift = 0.737), coinciding with the disclosure and exploitation of CVE-2023-46805 and CVE-2024-21887 affecting Ivanti Connect Secure VPN appliances. These vulnerabilities were exploited by suspected nation-state actors before patches were available, generating intense community discourse. DGT detects the shift as *ivanti* migrated from a neutral vendor reference to a shared shorthand for enterprise VPN zero-day exploitation, co-occurring with terms like *zero-day*, *unauthenticated*, *rce*, *lateral*, and *nation-state*. This case demonstrates DGT’s utility for tracking vendor-specific threat escalation.

6.3.6 Jailbreak — iOS to LLM (Peak S10→S11, Shift = 0.610)

The *jailbreak* trajectory is arguably the most theoretically significant case in the corpus: a *cognitive frame transfer* in which the community repurposes an existing concept—bypassing a vendor’s intended system restrictions—for a qualitatively new class of target. Through Spells 1–9, *jailbreak* referred almost exclusively to iOS and Android firmware bypass: techniques for

removing manufacturer restrictions from mobile devices. As shown in Figure 2, the shift trajectory for *jailbreak* is moderate but gradually rising through this period—particularly from Spell 7 onward as language models began attracting community attention—reaching its peak at S10→S11 (shift = 0.610). From Spell 10 onward, *jailbreak*’s semantic neighbourhood reorganises entirely around LLM safety-filter circumvention: *prompt*, *injection*, *chatgpt*, *llm*, *bypass*, *guardrail*. The same cognitive structure—defeating a vendor’s restriction mechanism—has been applied to an entirely new system class. DGT detected this repurposing in real time as ChatGPT entered the threat-actor toolkit, providing the intelligence signal for an emerging attack surface that now occupies a significant share of community discourse.

6.4 Forward-Looking Watchlist

Table 10 presents the DGT Watchlist for the forthcoming spell—the top annotated terms by predicted next-spell cosine shift, filtered through the curated security annotation dictionary. The EMA extrapolation is intentionally parsimonious: it projects the most recent trajectory forward without assuming mean-reversion, so a term with a persistently rising shift trajectory receives a higher watchlist score than one with the same cumulative average but a declining recent trend. The watchlist therefore carries genuine forecasting signal beyond a simple recency lookback, though the signal is extrapolative rather than structural.

The watchlist reveals several actionable intelligence patterns. The top-ranked terms *woocommerce* and *shortcode* point toward a forthcoming intensification in WordPress plugin supply-chain discourse, consistent with recent patterns of critical access-control and code-injection vulnerabilities in widely deployed CMS ecosystems. The appearance of *authorization* at rank 3 tracks the dominant OWASP Top 10 vulnerability class (A01: Broken Access Control), suggesting

Table 10: DGT Forward-Looking Watchlist: Top Predicted Next-Spell Semantic Shifts			
Term	Pred. Shift	Recent Shift	Security Interpretation
woocommerce	0.475	0.222	WordPress/WooCommerce plugin supply-chain vulnerabilities
shortcode	0.410	0.098	WordPress shortcode injection — server-side code execution
authorization	0.389	0.207	Broken access-control vulnerabilities (OWASP A01)
xwiki	0.364	0.254	Wiki-platform SSTI and RCE vulnerability pattern
over-read	0.355	0.303	Out-of-bounds read (information disclosure, ASLR bypass)
mcafee	0.337	0.090	Endpoint-security driver exploitation (kernel privilege)
low-privileg	0.336	0.207	Local privilege-escalation exploit chains
darkweb	0.336	0.530	Dark-web marketplace activity and data brokering
jailbreak	0.335	0.092	LLM safety-bypass technique — iOS→AI drift continues
prompt	0.327	0.311	Prompt injection and AI input manipulation techniques
<i>Note: Predicted shift is the DGT EMA-forecast cosine distance for the forthcoming spell (Spell 13). Recent shift is the observed cosine distance at the most recent transition (S11→S12). Terms filtered through a curated security annotation dictionary. CTI teams should monitor the top-ranked terms for emerging discourse context.</i>			

continued attacker focus on privilege and authentication bypass rather than memory corruption. The continued predicted shift for *jailbreak* (rank 9, predicted = 0.335, recent = 0.092) confirms that LLM safety-bypass techniques remain semantically evolving—the community is developing new vocabulary faster than it stabilises, an early signal of ongoing adversarial AI research. A CTI analyst receiving this watchlist would be advised to increase collection monitoring around WordPress ecosystem vulnerabilities, low-privilege escalation chains, and prompt injection tooling in the forthcoming observational period.

7 DISCUSSION

7.1 Theoretical Contributions

This paper makes three theoretical contributions to the IS research literature.

A Novel DSR Artifact Pair. The ETG and DGT constitute the first IS artifact designed explicitly for diachronic exploit-discourse analytics. The ETG extends the graph-of-words representation (Mihalcea & Tarau, 2004; Rousseau & Vazirgiannis, 2015) to the cybersecurity domain with a cumulative temporal design grounded in the theory of knowledge accumulation in communities of practice. The DGT extends the graph transformer paradigm (Kreuzer et al., 2021; Rampášek et al., 2022) with three novel components: adjacency-constrained self-attention across co-occurring exploit terms, a recency-weighted EMA temporal context loss that enforces smooth embedding trajectories, and post-hoc EMA shift extrapolation for next-spell prediction—together operationalising the four Design Science principles for the CTI use case. Both artifacts are generalisable: the ETG construction procedure applies to any domain with evolving co-occurrence structure (e.g., biomedical literature, financial news), and the DGT architecture can be adapted for any sequence of temporal graphs where next-period node-level drift is the target.

Empirically Validated Design Principles. All four design principles (DP1–DP4) are validated through the ablation study, contributing to the IS design knowledge base (Gregor & Hevner, 2013). DP1 is validated by the cumulative-ETG ablation ($\Delta = +0.042$), confirming that accumulated co-occurrence history is essential. DP2 is validated by two ablations: removing the within-spell adjacency mask ($\Delta = +0.054$) confirms that constraining attention to co-occurrence neighbours is functionally necessary, and removing Laplacian positional encodings ($\Delta = +0.109$)—the largest DP2 effect—confirms that spectral structural fingerprints are the most critical component for discriminating high-shift terms. DP3 is validated by showing that removing the temporal context loss ($\Delta = +0.034$) or removing the temporal training sequence entirely ($\Delta = +0.116$, the largest single ablation effect) both degrade performance significantly, confirming that temporally ordered training—not graph transformer architecture alone—drives predictive performance. DP4

is validated through watchlist content analysis: restricting the corpus to the hacker-community layer alone reduces Top-50 watchlist overlap with DGT-full to just 3 of 50 terms (6%), with the single-source model surfacing Turkish defacement handles and forum pseudonyms as its top predicted shifters in place of the security-relevant vocabulary (WooCommerce, authorisation bypass, vendor names) that multi-source enrichment provides. Together, these findings provide actionable design guidance for future IS artifacts in the temporal knowledge representation domain.

Lexical Threat Intelligence Lag: Formal Definition and Measurement. We introduce the construct of *lexical threat intelligence lag*, defined as follows.

Definition. Let w be an exploit-discourse term tracked by DGT. The *lexical threat intelligence lag* for w , $\Delta(w)$, is the interval in months between (1) $T_{\text{DGT}}(w)$: the end date of spell k at which w first appears in DGT’s top- N predicted-shift watchlist for the $k \rightarrow k+1$ transition (i.e., the date a CTI analyst would receive the watchlist alert), and (2) $T_{\text{formal}}(w)$: the earliest formal cataloguing of the associated threat, defined as the CISA KEV addition date if available, otherwise the first NVD CVE publication for the associated attack family.

$$\Delta(w) = T_{\text{formal}}(w) - T_{\text{DGT}}(w)$$

Positive Δ indicates that the DGT watchlist surfaced w before formal cataloguing (*intelligence lead*). Negative Δ indicates that formal cataloguing preceded the DGT peak-shift detection (*post-disclosure stabilisation*). If no formal cataloguing event exists for the threat associated with w , Δ is undefined and DGT’s detection represents the only available structured intelligence signal (*paradigm shift*).

Table 11 applies this measurement to the six case-study terms using the spell boundaries from

Table A1. $T_{\text{DGT}}(w)$ is the end date of the spell immediately before the peak shift transition; $T_{\text{formal}}(w)$ is sourced from publicly available CISA KEV and NVD publication records.

Table 11: Lexical Threat Intelligence Lag Across Six Case Study Terms					
Term	Peak Trans.	T_{DGT}	T_{formal}	Δ (mo.)	Lag type
<i>iot</i>	S01→S02	Nov 2016	Oct 2016 ^a	−1	Concurrent
<i>rce</i>	S02→S03	Sep 2017	Mar 2017 ^b	−6	Post-disclosure stab.
<i>ransomware</i>	S04→S05	Jun 2019	—	N/A	Paradigm shift
<i>stealer</i>	S06→S07	Feb 2021	—	N/A	Paradigm shift
<i>ivanti</i>	S09→S10	Sep 2023	Jan 2024 ^c	+4	Intel. lead
<i>jailbreak</i>	S10→S11	Aug 2024	—	N/A	Paradigm shift
Note: T_{DGT} = end date of spell k at which DGT watchlist would flag term w for the $k \rightarrow k+1$ transition. T_{formal} = earliest CISA KEV or NVD CVE date for the associated threat. $\Delta = T_{\text{formal}} - T_{\text{DGT}}$; positive = DGT led formal cataloguing. “—” = no CVE/KEV cataloguing event for this threat class; DGT is the only structured signal. Sources: ^a US-CERT TA16-288A (Oct 14, 2016). ^b CVE-2017-0144 (NVD, Mar 14, 2017). ^c CISA KEV entry for CVE-2023-46805 (Jan 10, 2024).					

The six terms yield a typology of three lag patterns. *Concurrent or post-disclosure stabilisation* ($\Delta \leq 0$): for Mirai and EternalBlue/WannaCry, the formal alert or CVE was published before or at the same time as DGT’s peak detection. Critically, this does not mean DGT is uninformative in these cases—the negative Δ for *rce* ($\Delta = -6$ months) reflects that CVE-2017-0144 was published in March 2017, but the semantic reorganisation of hacker discourse around “rce” as the dominant exploit-class label for the WannaCry era occurred over the subsequent six months, as the community integrated EternalBlue into its conceptual framework. DGT detects this *lexical adoption* event—when a threat concept becomes the community’s settled vocabulary—which formal CVE databases never record. *Intelligence lead* ($\Delta > 0$): for Ivanti, DGT’s S09 watchlist (available September 2023) surfaced *ivanti* as a predicted high-shift term four months before CISA added CVE-2023-46805 to the KEV catalogue (January 10, 2024)—the period during which exploitation was active in the wild but not yet formally acknowledged. *Paradigm shift* (Δ undefined): for Ransomware-as-a-Service, infostealer markets, and LLM jailbreaking, no CVE or KEV cataloguing event exists for the threat *class* as opposed to individual exploits; DGT is the only systematic mechanism for detecting when these paradigms achieve community-wide vocabulary

saliency.

Two testable propositions follow from this typology, directing future empirical work:

Proposition 1 (Intelligence-lead hypothesis). For emerging vulnerabilities that are actively exploited in the wild before formal public disclosure (zero-day scenarios), $\Delta(w) > 0$: DGT’s watchlist provides positive lead time over CISA KEV cataloguing. (Empirically illustrated: Ivanti, $\Delta = +4$ months.)

Proposition 2 (Paradigm-shift hypothesis). For attack-class terms representing new criminal business models or technology repurposing—rather than specific exploitable vulnerabilities— $T_{\text{formal}}(w)$ is undefined, and DGT’s semantic shift detection represents the primary structured intelligence signal unavailable from CVE or KEV sources.

Together, these propositions position lexical threat intelligence lag as a measurable, falsifiable construct that grounds the IS cybersecurity analytics agenda in an empirically tractable research programme rather than an informal observation. The construct is computable from any deployment of the ETG/DGT pipeline against a vulnerability-adjacent corpus with access to CVE and KEV publication dates.

7.2 Practical Implications

For **CTI analysts** and security operations centre (SOC) practitioners, the DGT Watchlist provides a proactive alerting mechanism that complements existing signature-based and frequency-based threat feeds. Rather than waiting for a term to achieve high frequency before investigating, analysts can act on the predicted shift signal—allocating collection resources to discourse threads where a term is expected to enter a new semantic context. The six-term case study demonstrates that such

shifts correspond to meaningful, practitioner-relevant threat-landscape changes.

For **Chief Information Security Officers (CISOs)** and security programme managers, the temporal shift forecasts provide a strategic planning horizon. Terms predicted to shift in the forthcoming spell often correspond to attack surfaces that will receive increased community attention—and potentially increased exploit development—in the subsequent 9–18 months. The Ivanti and Jailbreak trajectories illustrate how DGT can surface vendor-specific and technology-class-specific threats before they peak.

For **government agencies and ISACs**, the DGT’s output can augment the CISA KEV advisory process. The lexical threat intelligence lag analysis (Table 11) shows that for zero-day exploitation scenarios—the highest-priority KEV cases— $\Delta(w)$ can be positive: the Ivanti case demonstrates a 4-month DGT lead over the KEV listing date. Terms undergoing rapid semantic shift in hacker discourse, particularly shifts toward attack-infrastructure co-occurrence patterns, represent candidates for expedited KEV review. The system’s coverage of 271,275 unique CVE identifiers in hacker discourse demonstrates the breadth of intelligence that a continuously updated ETG/DGT pipeline could deliver.

From a **technical deployment** perspective, the DGT pipeline is accessible to organisations with moderate computational resources: the full 12-spell pipeline from corpus ingestion through watchlist generation runs on a single CUDA GPU within a standard workday of compute.

7.3 Limitations and Future Research

Several limitations of the current study warrant acknowledgement. First, HackerSignal is an English-language corpus; multilingual hacker communities—particularly active in Eastern European, Chinese, and Arabic-language forums—are not represented. Future work should apply

cross-lingual transfer learning (Ebrahimi et al., 2022) to extend DGT to multilingual exploit discourse. Second, the ETG edge cap (300,000 per spell) may truncate rare but important co-occurrences involving low-frequency but high-specificity attack terms. Dynamic edge selection strategies, such as retaining edges by mutual information rather than raw count, are a promising avenue. Third, $R^2 = 0.045$ is low, reflecting the heavy-tailed shift distribution in which many terms exhibit near-zero drift; R^2 is not an appropriate primary metric for this distribution. We therefore report both MAE and Spearman ρ as co-primary evaluation criteria throughout the benchmark (Tables 4–7). Future work should consider additionally adopting NDCG for position-weighted rank evaluation of the CTI watchlist. Fourth, BERT-based contextual encoders were not benchmarked in this study due to the distinct fine-tuning infrastructure required for graph-level tasks; a thorough comparison with domain-specific models (CTI-BERT, SecBERT) is reserved for future work. Fifth, while the primary benchmark evaluation uses a single held-out spell (Spell 12), we conducted a five-fold walk-forward validation (training on Spells 1– k , evaluating on the held-out transition for $k \in \{7, 8, 9, 10, 11\}$) to confirm temporal stability. DGT achieves a mean walk-forward MAE of 0.060 ± 0.008 across all five folds with Top-50 overlap of 1.00 in every fold, and significantly outperforms all five baselines tested (PPMI, SGNS, GloVe, DeepWalk, frequency slope) on at least three of five paired t -tests ($p < 0.05$).

8 CONCLUSION

The cybersecurity threat landscape is not static, and the vocabulary through which hacker communities articulate it is not static either. Terms shift meaning as attack techniques mature, as new tools achieve commercial scale, as geopolitical actors adopt novel targets, and as community

attention pivots to new technology classes. The Exploit Text Graph and Diachronic Graph Transformer, presented here as a Design Science Research artifact pair, operationalise the detection and prediction of these semantic shifts at scale.

Over a decade of hacker forum discourse—1,783,276 posts across 12 temporal spells—DGT produces shift predictions that significantly outperform all 36 of 36 tested external benchmark models at FDR $q < 0.001$, including all diachronic semantic methods, graph embedding models, dynamic graph neural networks, and modern graph transformers. Four design principles, validated through ablation, establish that cumulative graph encoding, a recency-weighted temporal context loss, adjacency-constrained self-attention, and multi-source corpus construction are individually necessary and jointly sufficient conditions for effective diachronic exploit analytics. A five-fold walk-forward validation confirms temporal stability with a mean MAE of 0.060 across five forecasting horizons and a perfect Top-50 overlap in every fold. A six-term case study anchored to verifiable real-world cybersecurity events—from the Mirai botnet and WannaCry through Ransomware-as-a-Service emergence (the single highest cosine shift in the corpus, 1.117) and LLM jailbreaking—demonstrates that DGT’s shift detections are not statistical artefacts but reflect genuine, practitioner-relevant changes in the threat landscape.

The formally operationalised construct of lexical threat intelligence lag—defined as the interval $\Delta(w) = T_{\text{formal}}(w) - T_{\text{DGT}}(w)$ between DGT watchlist detection and CISA KEV or CVE publication—provides a measurable, falsifiable lens for IS cybersecurity analytics research. Applied to the six case-study terms, the construct yields three lag patterns (concurrent, post-disclosure stabilisation, intelligence lead) and surfaces the Ivanti zero-day as a concrete instance of $\Delta = +4$ months lead time over formal cataloguing. Future work should extend DGT to multilingual corpora, conduct systematic lag measurement across all 271,275 CVEs observed in the

corpus, and conduct formal user studies with CTI analysts to quantify the operational value of the watchlist as an intelligence product. We invite the IS cybersecurity analytics community to adopt the ETG and DGT as open research artifacts and to build upon the design principles and theoretical constructs established here.

REFERENCES

- Ampel, B. M., & Samtani, S. (2026). HackerSignal: A large-scale multi-source dataset linking hacker community discourse to the CVE vulnerability lifecycle [NeurIPS 2026 Datasets and Benchmarks Track submission. Dataset available at <https://huggingface.co/datasets/BenAmpel/HackerSignal>]. <https://huggingface.co/datasets/BenAmpel/HackerSignal>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- CISA. (2024). *Known exploited vulnerabilities catalog* (tech. rep.). Cybersecurity and Infrastructure Security Agency. <https://www.cisa.gov/known-exploited-vulnerabilities-catalog>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd). Lawrence Erlbaum Associates.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 4171–4186. <https://doi.org/10.18653/v1/N19-1423>
- Ebrahimi, M., Nunamaker, J. F., & Chen, H. (2022). Semi-supervised cyber threat identification in dark web forums: A zero-shot learning with adversarial autoencoder approach. *Journal of Management Information Systems*, 39(2), 331–364. <https://doi.org/10.1080/07421222.2022.2063551>
- Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2), 337–355. <https://doi.org/10.25300/MISQ/2013/37.2.01>
- Grover, A., & Leskovec, J. (2016). Node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855–864. <https://doi.org/10.1145/2939672.2939754>
- Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016). Diachronic word embeddings reveal statistical laws of semantic change. *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, 1489–1501. <https://doi.org/10.18653/v1/P16-1141>
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30.
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105. <https://doi.org/10.2307/25148625>
- Jacobs, J., Romanosky, S., Adjerid, I., & Baker, W. (2021). Exploit prediction scoring system (EPSS). *Digital Threats: Research and Practice*, 2(3), 1–17. <https://doi.org/10.1145/3436242>
- Kim, J., Nguyen, D., Min, S., Cho, S., Lee, M., Lee, H., & Hong, S. (2022). Pure transformers are powerful graph learners. *Advances in Neural Information Processing Systems*, 35, 14582–14595.
- Kim, Y., Chiu, Y.-I., Hanaki, K., Hegde, D., & Petrov, S. (2014). Temporal analysis of language through neural language models. *Proceedings of the ACL 2014 Workshop on Language*

- Technologies and Computational Social Science*, 61–65. <https://doi.org/10.3115/v1/W14-2517>
- Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations*.
- Kreuzer, D., Beaini, D., Hamilton, W., Létourneau, V., & Tossou, P. (2021). Rethinking graph transformers with spectral attention. *Advances in Neural Information Processing Systems*, 34, 21618–21629.
- Kulkarni, V., Al-Rfou, R., Perozzi, B., & Skiena, S. (2015). Statistically significant detection of linguistic change, 625–635. <https://doi.org/10.1145/2736277.2741627>
- Kumar, S., Zhang, X., & Leskovec, J. (2019). Predicting dynamic embedding trajectory in temporal interaction networks. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1269–1278. <https://doi.org/10.1145/3292500.3330895>
- Mihalcea, R., & Tarau, P. (2004). TextRank: Bringing order into text, 404–411.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, 26, 3111–3119.
- Nunes, E., Diab, A., Gunn, A., Marin, E., Mishra, V., Paliath, V., Robertson, J., Shakarian, J., Thart, A., & Shakarian, P. (2016). Darknet and deepnet mining for proactive cybersecurity threat intelligence, 7–12. <https://doi.org/10.1109/ISI.2016.7745435>
- Pareja, A., Domeniconi, G., Chen, J., Ma, T., Suzumura, T., Kanezashi, H., Kaler, T., Schardl, T. B., & Leiserson, C. E. (2020). EvolveGCN: Evolving graph convolutional networks for dynamic graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34, 5363–5370. <https://doi.org/10.1609/aaai.v34i04.5988>
- Peffer, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>
- Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). Deepwalk: Online learning of social representations. *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 701–710. <https://doi.org/10.1145/2623330.2623732>
- Rampášek, L., Galkin, M., Dwivedi, V. P., Luu, A. T., Wolf, G., & Beaini, D. (2022). Recipe for a general, powerful, scalable graph transformer. *Advances in Neural Information Processing Systems*, 35, 14501–14515.
- Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. (2020). Temporal graph networks for deep learning on dynamic graphs. *arXiv preprint arXiv:2006.10637*.
- Röttger, P., & Pierrehumbert, J. B. (2021). Temporal adaptation of BERT and performance on downstream document classification: Insights from social media. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2400–2412. <https://doi.org/10.18653/v1/2021.findings-emnlp.206>
- Rousseau, F., & Vazirgiannis, M. (2015). Main core retention on graph-of-words for single-document keyword extraction, 382–393. https://doi.org/10.1007/978-3-319-16354-3_42
- Samtani, S., Chinn, R., Chen, H., & Nunamaker, J. F. (2017). Exploring emerging hacker assets and key hackers for proactive cyber threat intelligence. *Journal of Management Information Systems*, 34(4), 1023–1053. <https://doi.org/10.1080/07421222.2017.1394049>

- Samtani, S., Yu, S., Zhu, H., Patton, M., Lacity, M., & Chen, H. (2020). Identifying SCADA systems and their vulnerabilities on the internet of things: A text-mining approach. *IEEE Intelligent Systems*, 35(1), 14–22. <https://doi.org/10.1109/MIS.2019.2942018>
- Samtani, S., Zhu, H., & Chen, H. (2022). Linking exploits from the dark web to known vulnerabilities for proactive cyber threat intelligence. *MIS Quarterly*, 46(2), 911–946. <https://doi.org/10.25300/MISQ/2022/15392>
- Sankar, A., Wu, Y., Gou, L., Zhang, W., & Yang, H. (2020). DySAT: Deep neural representation learning on dynamic graphs via self-attention networks. *Proceedings of the 13th International Conference on Web Search and Data Mining*, 519–527. <https://doi.org/10.1145/3336191.3371845>
- Tang, J., Qu, M., Wang, M., Zhang, M., Yan, J., & Mei, Q. (2015). LINE: Large-scale information network embedding. *Proceedings of the 24th International Conference on World Wide Web*, 1067–1077. <https://doi.org/10.1145/2736277.2741093>
- Tavabi, N., Goyal, P., Almukaynizi, M., Shakarian, P., & Lerman, K. (2018). Darkembed: Exploit prediction with neural language models. 32, 7849–7856.
- Ullman, T. D., & Samtani, S. (2024). Enhancing vulnerability prioritization in cloud computing with multi-view learning. *MIS Quarterly*, 48(1), 1–26. <https://doi.org/10.25300/MISQ/2024/17408>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. *International Conference on Learning Representations*.

APPENDIX A: FULL EXPLOIT TEXT GRAPH GROWTH STATISTICS

Table A1 presents the complete ETG growth statistics for all 12 temporal spells.

Table 12: Table A1. Full ETG Growth Statistics Across All 12 Temporal Spells						
Spell	Date Range	Posts	Nodes	Edges	Co-occ. Wt (M)	CVEs
S01	Jan 2016 – Nov 2016	330,404	20,642	280,344	11.6	6,019
S02	Jan 2016 – Sep 2017	627,829	22,654	300,000	21.6	20,911
S03	Jan 2016 – Jul 2018	704,322	23,210	300,000	24.1	37,172
S04	Jan 2016 – Jun 2019	774,023	24,526	300,000	26.6	50,748
S05	Jan 2016 – Apr 2020	926,846	25,517	300,000	30.5	68,812
S06	Jan 2016 – Feb 2021	998,928	26,893	300,000	33.4	84,867
S07	Jan 2016 – Jan 2022	1,052,547	27,627	300,000	36.0	103,672
S08	Jan 2016 – Nov 2022	1,100,924	27,994	300,000	38.4	126,118
S09	Jan 2016 – Sep 2023	1,140,184	28,372	300,000	40.5	152,479
S10	Jan 2016 – Aug 2024	1,190,949	28,573	300,000	43.7	185,289
S11	Jan 2016 – Jun 2025	1,250,508	28,790	300,000	47.3	224,173
S12	Jan 2016 – Apr 2026	1,783,276	30,000	300,000	55.7	271,275
Note: Edges are capped at 300,000 per spell (top co-occurrence-weighted edges retained). Co-occurrence weight is the total sum of all edge weights in the spell. CVEs Observed is the count of distinct CVE identifiers appearing in hacker-forum posts up to and including that spell.						

APPENDIX B: DGT ARCHITECTURE NOTATION

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{W}_t)$ denote the Exploit Text Graph at spell t , with $N = |\mathcal{V}| = 6,000$ tracked nodes. The 9-dimensional base node feature matrix is $\mathbf{X}_t \in \mathbb{R}^{N \times 9}$ (log-transformed frequency and degree statistics plus three morphological flags). Laplacian positional encodings are the $k = 16$ eigenvectors of the *normalised* graph Laplacian $\mathbf{L}_t = \mathbf{I} - \mathbf{D}_t^{-1/2} \mathbf{A}_t \mathbf{D}_t^{-1/2}$ corresponding to the k smallest non-trivial eigenvalues, $\square_t \in \mathbb{R}^{N \times 16}$, giving augmented features $\tilde{\mathbf{X}}_t = [\mathbf{X}_t || \square_t] \in \mathbb{R}^{N \times 25}$.

The adjacency-masked transformer encoder $f_\theta : \mathbb{R}^{N \times 25} \times \{0, 1\}^{N \times N} \times \mathbb{Z} \rightarrow \mathbb{R}^{N \times d}$ (with $d = 128$, 4 heads, 2 layers, GELU activation, 0.2 dropout) processes each spell independently. For spell t , the input projection adds a learned spell embedding $\mathbf{s}_t$:

$$\mathbf{H}_t^{(0)} = \mathbf{W}_{\text{in}} \tilde{\mathbf{X}}_t + \mathbf{1}_N \mathbf{s}_t^\top, \quad \mathbf{s}_t = \text{Embedding}(t) \in \mathbb{R}^d \quad (12)$$

Self-attention is masked by the ETG adjacency: non-neighbours receive $-\infty$ attention logits. After

two transformer layers, a residual bypass with learnable scalar $\alpha = \sigma(\theta)$ (with θ initialised to 0, so $\alpha_0 = 0.5$) blends transformer output and input:

$$\mathbf{Z}_t = \alpha f_{\text{enc}}(\mathbf{H}_t^{(0)}) + (1 - \alpha) \mathbf{H}_t^{(0)} \quad (13)$$

Per-spell embeddings are produced by a two-layer MLP projection followed by L2 normalisation:

$$\mathbf{E}_t = \text{L2}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{Z}_t)) \in \mathbb{R}^{N \times d} \quad (14)$$

The composite training loss is:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{link}} + \lambda \mathcal{L}_{\text{temporal}}, \quad \lambda = 0.1 \\ \mathcal{L}_{\text{link}} &= - \sum_{t=1}^T \sum_{(u,v)} [y_{uv} \log \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}) + (1 - y_{uv}) \log(1 - \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}))] \\ \mathcal{L}_{\text{temporal}} &= \sum_{t=2}^T \|\mathbf{E}_t - \bar{\mathbf{E}}_{<t}\|_F^2, \quad \bar{\mathbf{E}}_{<t} = \sum_{s<t} w_s \mathbf{E}_s, \quad w_s \propto e^{-0.5(t-1-s)} \end{aligned} \quad (15)$$

where $y_{uv} \in \{0, 1\}$ is the edge label and $\bar{\mathbf{E}}_{<t}$ is the recency-weighted EMA of all past spell embeddings (detached from the computation graph). Training uses AdamW (lr 10^{-3} , weight decay 10^{-4}) with 10% linear warmup followed by cosine annealing to 10^{-5} ; gradients are clipped to unit norm.

Shift prediction is post-hoc (no learned prediction head). Observed cosine shifts $c_{v,t} = 1 - \cos(\mathbf{e}_{v,t-1}, \mathbf{e}_{v,t})$ are computed over the training window for each term v , and the predicted next-spell

shift is the recency-weighted EMA with decay $\gamma = 0.7$:

$$\hat{y}_v = \sum_{t=2}^T w_t c_{v,t}, \quad w_t \propto \gamma^{T-t}, \quad \hat{y}_v = \max(0, \hat{y}_v) \quad (16)$$

APPENDIX C: FULL BENCHMARK LEADERBOARD

Table C1 reports all 38 benchmark models with complete evaluation metrics, grouped by experiment tier. NP denotes null predictor (near-zero MAE achieved by predicting zero shift for all terms). All delta values compare to DGT MAE = 0.077.

Table 13: Table C1. Full Benchmark Leaderboard (38 Models)					
Model	MAE	RMSE	Top-50	Δ	Sig.
<i>Experiment 1: Non-Neural Classic</i>					
GloVe Log-Cooccurrence SVD	0.121	0.204	0.00	+0.044	Yes
PPMI + SVD + Procrustes	0.132	0.212	0.06	+0.055	Yes
SGNS Shifted PPMI SVD	0.132	0.211	0.02	+0.055	Yes
Kleinberg Burst Approx.	1.313	1.738	0.10	+1.236	Yes
ARIMA Trend (NP)	0.000	0.000	0.04	−0.077	†
Degree Trend (NP)	0.000	0.000	0.02	−0.077	†
ETS Trend (NP)	0.000	0.000	0.04	−0.077	†
TF-IDF Slope (NP)	0.000	0.000	0.06	−0.077	†
PageRank Trend (NP)	0.000	0.000	0.04	−0.077	†
Raw Frequency Slope (NP)	0.000	0.000	0.04	−0.077	†
Weighted Degree Trend (NP)	0.000	0.000	0.02	−0.077	†
<i>Experiment 2: Graph Embedding and Dynamic GNN</i>					
DeepWalk SVD	0.111	0.207	0.00	+0.033	Yes
Node2Vec Biased Walk	0.121	0.203	0.02	+0.044	Yes
LINE First Order	0.149	0.244	0.00	+0.072	Yes
LINE Second Order	0.150	0.271	0.02	+0.073	Yes
GAT Link Reconstr. (NP)	0.031	0.108	0.00	−0.046	†
GCN Link Reconstr. (NP)	0.032	0.117	0.04	−0.045	†
GraphSAGE Link Reconstr. (NP)	0.030	0.103	0.06	−0.047	†
CAWN Causal Walk	0.102	0.167	0.04	+0.025	Yes
DySAT Structural Temporal	0.111	0.175	0.04	+0.034	Yes
EvolveGCN Temporal Smooth	0.086	0.154	0.02	+0.009	Yes
GraphMixer Event Mix	0.115	0.179	0.02	+0.037	Yes
JODIE Projected Drift	0.061	0.135	0.00	−0.016	No
TGAT Time-Weighted Edges	0.119	0.188	0.02	+0.041	Yes
TGN Memory Smooth	0.074	0.146	0.02	−0.003	No
<i>Experiment 3: Modern Graph Transformers</i>					
GraphGPS LapPE Proxy (NP)	0.030	0.103	0.06	−0.047	†
TokenGT Graph Token SVD	0.138	0.250	0.02	+0.061	Yes
<i>Experiment 4: Ablations</i>					
No Cumulative ETG	0.149	0.244	0.00	+0.072	Yes
k -Precedence Window ($k = 4$)	0.149	0.244	0.00	+0.072	Yes
Same-Post Co-occurrence	0.149	0.244	0.00	+0.072	Yes
Direct Precedence Window	0.124	0.206	0.00	+0.047	Yes
Unweighted Edges	0.124	0.206	0.00	+0.047	Yes
No Masked Attention (NP)	0.000	0.000	0.00	−0.077	†
No Laplacian PE (NP)	0.053	0.120	0.26	−0.025	†
No Temporal Attention	0.132	0.212	0.06	+0.055	Yes
Static GT Only (NP)	0.036	0.059	0.26	−0.041	†
DGT Full (proposed)	0.077	0.178	1.00	—	—
<i>Note: MAE and RMSE rounded to 3 decimal places. NP = null predictor (Top-50 overlap ≤ 0.06 and MAE < 0.03). Sig. = DGT significantly better at FDR $q < 0.05$ († = null predictor, not applicable). All models trained/evaluated on the same 6,000-term vocabulary and same held-out Spell 12 shift target.</i>					